Increasing ratios in finite nonnegative linearization

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Abstract. For finite stochastic three-term recurrences, the companion paper [1] proves nonnegative linearization (property (P)) when the ratio of the diagonal to the downward coefficients is constant and the Newton coefficients of the profile $d_i = b_i + c_i$ are log-concave without internal zeros. We admit ratios $\rho(i) = c_i/d_i$ that increase with the index. Property (P) holds when the Newton coefficients of c are nonnegative and log-concave without internal zeros and those of b are those of c times nonincreasing nonnegative weights. On profiles with log-concave Newton coefficients and nondecreasing b it holds when ρ is linear or a convex quadratic in the index, when the Newton coefficients of ρ are nonnegative and log-concave without internal zeros, in any degree, and for cubic ratios $r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ under the single inequality $r\sigma \le \frac{7}{4}t\tau$. On profiles $\nu_1 i + \nu_2\binom{i}{2}$ with $\nu_1 > 0$ and $\nu_2 \ge 0$, ratios of any degree are admitted when the Newton coefficients of c are log-concave without internal zeros and b is nondecreasing. The proofs resum the moves of a kernel recurrence; the key inequalities for log-concave and cubic ratios are proved with exact computer certificates. A self-contained proof that log-concavity of Newton coefficients is preserved under pointwise products (Walkup's theorem) is included. Nondecreasing b alone, or nonnegative Newton coefficients of the ratio alone, do not suffice.

Keywords. Finite orthogonal polynomials; nonnegative linearization; log-concave sequences; Newton series; computer-assisted proof.

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1. Introduction

Szwarc [3, §5, Problem 2] asked for criteria, in the recurrence coefficients, for nonnegative linearization (property (P)) of finite systems of orthogonal polynomials. The companion paper [1] answers this for arrays whose diagonal and downward coefficients have a constant ratio: property (P) holds whenever the Newton coefficients of the profile are log-concave without internal zeros. This paper treats ratios $\rho(i) = c_i/(b_i + c_i)$ that increase with i.

Section 2 recalls the setting and the log-concavity inequality on which all proofs rest. Section 3 extends the kernel recurrence of [1] to arrays with independent diagonal and downward coefficients and proves property (P) when the Newton coefficients of c are log-concave without internal zeros and those of b are those of c times nonincreasing nonnegative

weights (Theorem 3.1 and Corollary 3.2), and for arrays that are quadratic in the index (Proposition 3.3). In general some moves of the recurrence have negative weight; Section 4 handles them by a resummation, for ratios that increase linearly (Theorem 4.1) or as a convex quadratic (Theorem 4.2) on every log-concave Newton profile, under the natural condition that b be nondecreasing. Section 5 treats ratios of higher degree: on profiles of Newton degree at most two (Theorem 5.1), and on every log-concave Newton profile when the Newton coefficients of the ratio are log-concave (Theorem 5.5) or, for cubic ratios, when $r\sigma \le t\tau$ (Theorem 5.6). Proposition 5.3, due to Walkup [4], shows that log-concavity of Newton coefficients is preserved under pointwise products. Section 6 records counterexamples and open problems.

The key inequalities of Theorems 5.5 and 5.6 are proved with exact computer certificates, and both theorems, together with the extension of Theorem 5.6 to $r\sigma \le \frac{7}{4}t\tau$, are also machine-verified; the other results have written proofs. The formalized statements are listed in the section on code and formalization.

2. Setting

We use the conventions of [1]. An array of real coefficients $(a_i, b_i, c_i)_{0\le i\le N}$ satisfies

$$\begin{gathered} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0), \end{gathered} \tag{2.1}$$

and $R_{-1} = 0$, $R_0 = 1$ and $xR_i = a_iR_{i+1} + b_iR_i + c_iR_{i-1}$ $(0 \le i < N)$ define its polynomials, with $R_i(1) = 1$. By [1, Lemma 2.1] the array has a positive orthogonality measure on $N+1$ real points, which include 1, and on this support $R_mR_n = \sum_k g(m,n;k)R_k$ with $g(m,n;k) = [R_m(J)]_{n,k}$, where J is the tridiagonal recurrence matrix; the array has *nonnegative linearization*, or *property (P)*, if all $g(m,n;k) \ge 0$. We write $d_i = b_i + c_i$ for the profile and, for $i \ge 1$ with $d_i > 0$, $\rho(i) = c_i/d_i$ for the ratio.

The Newton coefficients of a sequence $(u_i)_{0\le i\le N}$ are $\Delta^k u(0) = \sum_{i\le k}(-1)^{k-i}\binom{k}{i}u_i$, so that $u_n = \sum_k \binom{n}{k}\Delta^k u(0)$. A nonnegative, finitely supported sequence $(u_k)_{k\ge0}$ is *log-concave without internal zeros* if its positive entries form an interval and $u_k^2 \ge u_{k-1}u_{k+1}$ for all $k \ge 1$. For such a sequence put $f_n(\ell) = \sum_{h\ge0}\binom{\ell}{h}u_{n+h}$, and, for integers $p, q \ge 0$,

$$W_{p,q}(\ell, a) = \sum_{n\ge p} \binom{n-p+q}{q}\binom{a}{n} f_n(\ell),$$

with $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$ and $C_j = W_{2j+2,j+1}$ when these are formed from the Newton coefficients of the profile.

Lemma 2.1 *([1]).* *Let $(u_k)_{k\ge0}$ be nonnegative, finitely supported and log-concave without internal zeros. Then $(u_k + u_{k+1})_{k\ge0}$ has the same property, hence so does $(f_n(\ell))_{n\ge0}$ for every $\ell \ge 0$, and*

$$f_i(\ell)\, f_l(\ell+1) \ge f_0(\ell+1)\, f_{i+l}(\ell) \qquad (i, l \ge 1,\ \ell \ge 0).$$

Proof. For a nonnegative finitely supported sequence u that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_i u_{i+2}) + (u_i u_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(f_n(\ell))_n$ is obtained from $(u_n)_n$ by applying $u \mapsto v$ ℓ times. For such a sequence g, balanced products dominate: $g_i g_l \geq g_0 g_{i+l}$ and $g_i g_{l+1} \geq g_1 g_{i+l}$ for $i, l \geq 1$. Adding these with $g = f(\ell)$ and using $g_l + g_{l+1} = f_l(\ell+1)$ and $g_0 + g_1 = f_0(\ell+1)$ gives the claim. ∎

For the profile, with $u_k = \nu_k$ and $\nu_0 = 0$, one has $f_0(\ell) = d_\ell$, and the inequality reads $f_i(\ell) f_l(\ell+1) \geq d_{\ell+1} f_{i+l}(\ell)$. The paper [1] proves property (P) for constant ratios by expanding the product coefficients in a kernel $\omega_{\ell,a}$ on $\Omega = \{(\ell, a) : \ell, a \geq 0,\ \ell + a \leq N\}$, which satisfies a recurrence in ℓ whose moves have weights formed from the $W_{p,q}$, and by matching each move of odd index with two-step detours through moves of even index [1, Lemmas 3.2 and 3.4]. We extend that argument.

3. Ratios given by weights

The kernel argument also works in the stochastic normalization itself, with independent diagonal and downward coefficients. Let $a_i + b_i + c_i = 1$ for $0 \leq i \leq N$, with $b_0 = c_0 = 0$, $a_N = 0$, $a_i > 0$ for $i < N$ and $c_i > 0$ for $i \geq 1$. Write

$$\kappa_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} c_i, \qquad \beta_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} b_i \qquad (1 \leq k \leq N),$$

with $\kappa_k = \beta_k = 0$ for $k = 0$ and $k > N$, and put $\varphi_n(\ell) = \sum_{h \geq 0} \binom{\ell}{h} \kappa_{n+h}$ and $\psi_n(\ell) = \sum_{h \geq 0} \binom{\ell}{h} \beta_{n+h}$, so that $\varphi_0(\ell) = c_\ell$ and $\psi_0(\ell) = b_\ell$ for $\ell \leq N$.

Theorem 3.1 *(Varying ratio).* *Let (κ_k) be log-concave without internal zeros and $\beta_k \geq 0$ for all k. Suppose that for $0 \leq \ell \leq N - 1$*

$$\psi_1(\ell)\, \varphi_n(\ell+1) \geq c_{\ell+1}\, \psi_{n+1}(\ell) \quad (n \geq 2), \qquad \varphi_2(\ell)\, \psi_n(\ell+1) \geq c_{\ell+1}\, \psi_{n+2}(\ell) \quad (n \geq 1).$$

Then the system has nonnegative linearization.

Corollary 3.2. *If (κ_k) is log-concave without internal zeros and $\beta_k = w_k \kappa_k$ for $k \geq 1$ with $w_1 \geq w_2 \geq \cdots \geq 0$, then the system has nonnegative linearization. The ratio $c_i/(b_i + c_i)$ is nondecreasing in $i \geq 1$, and it is constant when w is constant on the support of κ; these are the arrays of [1, Theorem 3.1].*

For example, if c is a convex quadratic in i ($\kappa_1, \kappa_2 > 0$, $\kappa_k = 0$ for $k \geq 3$) and $b_i = w_1 \kappa_1 i + w_2 \kappa_2 \binom{i}{2}$ with $w_1 > w_2 \geq 0$, scaled so that $b_N + c_N = 1$, the ratio $c_i/(b_i + c_i)$ increases strictly. The hypotheses exclude the counterexample recorded in Section 6, whose diagonal coefficients $b = (0, 1, 1, 1)/91$ have Newton coefficients of both signs. For $N \geq 2$, a strictly increasing linear ratio on a linear profile violates the corollary's assumptions because $\Delta^2 b(0) < 0$; Proposition 3.3 covers it when b is nondecreasing.

Proof of Theorem 3.1. Put $e_i = \prod_{j<i} a_j/c_{j+1}$ and $E_i = e_i R_i$. Then $xE_i = c_{i+1}E_{i+1} + b_iE_i + a_{i-1}E_{i-1}$ on the support. The out-of-range terms are omitted: at the endpoints, $xE_0 = c_1E_1$ and $xE_N = b_NE_N + a_{N-1}E_{N-1}$. The series $F = \sum_i E_i z^i$ satisfies $xF = \mathcal{L}_z F$ with

$$\mathcal{L}_z = z^{-1}c(\theta_z) + b(\theta_z) + z\big(1 - b(\theta_z) - c(\theta_z)\big),$$

where $c(\theta) = \sum_k \kappa_k \binom{\theta}{k}$ and $b(\theta) = \sum_k \beta_k \binom{\theta}{k}$. As in [1, Section 3], $K_k = \sum_{m,n} L^k_{mn} z^m w^n$, with $E_mE_n = \sum_k L^k_{mn}E_k$, satisfies $(\mathcal{L}_z - \mathcal{L}_w)K_k = 0$, and $K_k = \sum \omega_{\ell,a} S^a Z^\ell$ with $\omega_{0,a} = \mathbf{1}_{a=k}$ and $\omega_{\ell,a} = 0$ for $\ell + a > N$. The computation in the proof of [1, Lemma 3.2] gives

$$\frac{(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell}{z-w} = S^aZ^\ell + \sum_{p\ge 0}\binom{a}{p} S^{a-p}Z^\ell\Big(\varphi_p(\ell)(\chi_{p-2} - \chi_p) + \psi_p(\ell)(\chi_{p-1} - \chi_p)\Big),$$

with $\chi_{-2} = -Z^{-1}$ and $\chi_{-1} = 0$. Collecting terms as there, with $W_{p,q}$ formed from φ and $W^\psi_{p,q}$ from ψ, gives

$$\begin{aligned} c_{\ell+1}\omega_{\ell+1,a} = a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\ge 0}(-1)^j\Big(&A^\psi_j(\ell-j, a+2j+1)\,\omega_{\ell-j,a+2j+1} + B_j(\ell-j,a+2j+2)\,\omega_{\ell-j,a+2j+2} \\ &+ \big(C_j + C^\psi_j\big)(\ell-j-1, a+2j+2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned}$$

where $A^\psi_j = W^\psi_{2j+1,j}$, $C^\psi_j = W^\psi_{2j+2,j+1}$, and $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$. The straight weight is $1 - c_{\ell+a} - b_{\ell+a} = a_{\ell+a}$, extended polynomially beyond N, and $\beta_0 = 0$. Up to this point no hypothesis of the theorem has been used: the recurrence holds for every array in the stochastic normalization. Under the hypotheses, all weights are nonnegative on Ω.

For $j \ge 1$, $0 \le \ell \le N-1$ and $a \ge 0$, the following detour inequalities hold, with weights having a negative second argument interpreted as zero:

$$\begin{aligned} A^\psi_0(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A^\psi_{j-1}(\ell+1,a-2) &\ge 4c_{\ell+1}A^\psi_j(\ell,a), \\ B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\ge 4c_{\ell+1}B_j(\ell,a), \\ B_0(\ell,a)\big(C_{j-1} + C^\psi_{j-1}\big)(\ell+1,a-2) &\ge 4c_{\ell+1}\big(C_j + C^\psi_j\big)(\ell,a). \end{aligned}$$

The proof of [1, Lemma 3.4] applies, with the lower bounds $A^\psi_0(\ell,a) \ge a\psi_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$. Lemma 2.1 for φ handles the products $\varphi_2(\ell)\varphi_n(\ell+1)$. The two hypotheses of the theorem handle $\psi_1(\ell)\varphi_n(\ell+1)$ for $n \ge 2$ and $\varphi_2(\ell)\psi_n(\ell+1)$ for $n \ge 1$, which are the indices that occur. The moves C_j and C^ψ_j have the same displacement and are treated as one move. With this, the proof of [1, Theorem 3.1] gives $\omega \ge 0$, hence $L^k_{mn} \ge 0$ and $g(m,n;k) = L^k_{mn}e_k/(e_me_n) \ge 0$. ∎

Proof of Corollary 3.2. Set $w_0 = w_1$; since $\kappa_0 = 0$, this leaves $\beta_0 = 0$. Fix ℓ and write $G_i = \varphi_i(\ell)$, $H_i = \psi_i(\ell)$ and $u_x = \binom{\ell}{x-i}$. For $i \ge 0$,

$$H_iG_{i+1} - H_{i+1}G_i = \sum_{x,y\ge i} u_xu_y\kappa_x\kappa_{y+1}(w_x - w_{y+1}).$$

Terms with $x \le y$ are nonnegative, and terms with $x = y+1$ vanish. The map $(x,y) \mapsto (y+1, x-1)$ is a bijection from the terms with $x \ge y+2$ onto those with

$i+1 \le x \le y$; the remaining terms have $x = i$. A term and its image sum to $\kappa_x\kappa_{y+1}(w_{y+1} - w_x)(u_{y+1}u_{x-1} - u_xu_y) \ge 0$, because w is nonincreasing and the zero-extended binomial coefficients are log-concave. Hence $H_{i+1}G_i \le H_iG_{i+1}$.

Since the support of (G_i) is an interval and $\beta_k = 0$ where $\kappa_k = 0$, chaining gives $H_{n+1}G_1 \le H_1G_{n+1}$, $H_{n+2}G_n \le H_nG_{n+2}$ and $H_{n+2}G_{n+1} \le H_{n+1}G_{n+2}$. Lemma 2.1 with $(i,l) = (1,n)$ and $(2,n)$ reads $G_1(G_n + G_{n+1}) \ge CG_{n+1}$ and $G_2(G_n + G_{n+1}) \ge CG_{n+2}$, where $C = G_0 + G_1 = c_{\ell+1}$ and $G_n + G_{n+1} = \varphi_n(\ell+1)$. Since $G_1 \ge \kappa_1 > 0$, $H_1(G_n + G_{n+1}) \ge CH_1G_{n+1}/G_1 \ge CH_{n+1}$. If $G_{n+2} > 0$, then $G_2(H_n + H_{n+1}) \ge G_2H_{n+2}(G_n + G_{n+1})/G_{n+2} \ge CH_{n+2}$; if $G_{n+2} = 0$, then $H_{n+2} = 0$. These are the two hypotheses of Theorem 3.1. The case $i = 0$ of the first display, with $\beta_0 = \kappa_0 = 0$, reads $b_{\ell+1}c_\ell \le b_\ell c_{\ell+1}$; for $1 \le \ell < N$, division by $c_\ell c_{\ell+1} > 0$ shows that b_i/c_i is nonincreasing and $c_i/(b_i + c_i)$ is nondecreasing. If w is constant on the support of κ, then b is proportional to c, and with $d = b + c$ the array is that of [1, Theorem 3.1]. ■

For arrays that are quadratic in the index, concave diagonal coefficients are allowed.

Proposition 3.3 *(Quadratic arrays).* *Let $b_n = \beta_1 n + \beta_2\binom{n}{2}$ and $c_n = \kappa_1 n + \kappa_2\binom{n}{2}$ $(0 \le n \le N)$ define an array as above. If $\kappa_2 \ge 0$, $\kappa_2 + \beta_2 \ge 0$ and $b_0 \le b_1 \le \cdots \le b_N$, then the system has nonnegative linearization.*

Proof. In the recurrence of the proof of Theorem 3.1, which holds for every array, $\varphi_n = \psi_n = 0$ for $n \ge 3$. So all moves with $j \ge 1$ vanish, and the moves with $j = 0$ have the weights $A_0^\psi(\ell,a) = \sum_{n\ge1}\binom{a}{n}\psi_n(\ell) = b_{\ell+a} - b_\ell$, $B_0(\ell,a) = \binom{a}{2}\kappa_2$ and $(C_0 + C_0^\psi)(\ell,a) = \binom{a}{2}(\kappa_2 + \beta_2)$. These and the straight weights $a_{\ell+a}$ are nonnegative on Ω, so $\omega \ge 0$ by induction on ℓ, and $g(m,n;k) \ge 0$ as in the proof of Theorem 3.1. ■

With the linear profile $d_n = n/N$ and the ratio $c_n/d_n = r + \kappa n/N$ $(1 \le n \le N,\ r > 0,\ \kappa \ge 0)$, one has $\kappa_2 = 2\kappa/N^2$ and $\kappa_2 + \beta_2 = 0$ for $N \ge 2$. For $N = 1$, property (P) follows directly from $R_1^2 = c_1R_0 + b_1R_1$ when $b_1 \ge 0$. So every linearly increasing ratio with a linear profile and nondecreasing b has property (P); b is nondecreasing exactly when $\kappa \le (1-r)N/(2N-1)$. Exact computations (`runs/varratio/quadratic.py`) show that each of the three hypotheses is needed: dropping any one of them gives quadratic arrays with a negative coefficient at $N \le 3$.

4. Linearly and quadratically increasing ratios

For curved profiles, linearly increasing ratios need not satisfy the hypotheses of Theorem 3.1: the Newton coefficients of b of order at least 3 can be negative, and then some moves of even index in the recurrence of its proof have negative weight and no two-step detour. A resummation of the positive moves handles this.

Theorem 4.1 *(Linearly increasing ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \le 1$ and $\kappa \ge 0$, and put $\rho_i = r + \kappa i/N$, $a_i = 1 - d_i$, $b_i = (1-\rho_i)d_i$ and $c_i = \rho_i d_i$. If b is nondecreasing, then*

the system has nonnegative linearization. The condition that b be nondecreasing is equivalent to $\kappa\big(N + d_{N-1}/(1-d_{N-1})\big) \le N(1-r)$.

For the linear profile $d_i = i/N$ the condition reads $\kappa \le \frac{N}{2N-1}(1-r)$, and these are arrays of Proposition 3.3. For a convex profile $d_{N-1} \le (N-1)/N$, so the admitted range of κ contains $[0, \frac{N}{2N-1}(1-r)]$. When $r < 1$, this containment is strict for a nonlinear profile. For example, $N = 4$, $d_i = (2^i - 1)/15$, $r = 1/2$ and $\kappa = 2/5$ are admitted, with $b = (0, 2/75, 3/50, 7/75, 1/10)$, although $(2N-1)\kappa/N = 7/10 > 1 - r$.

Proof. Write $s = 1 - r$ and $t = \kappa/N$, and let $f_n(\ell) = \Delta^n d(\ell)$ and $\nu_i = \Delta^i d(0)$ be as in Section 2, with $\nu_{-1} = \nu_0 = 0$. For $0 \le i < N$ put $R_i = d_i/f_1(i)$; then $f_1(i) \ge \nu_1 > 0$, and log-concavity of $(f_n(i))_{n\ge 0}$ gives $R_{i+1} - R_i = \big(f_1(i)^2 - d_i f_2(i)\big)/\big(f_1(i)f_1(i+1)\big) \ge 0$. Since $b_{i+1} - b_i = f_1(i)\big(s - t(i+1) - tR_i\big)$ and the bracket is nonincreasing in i, b is nondecreasing exactly when its last increment is nonnegative, that is, when $t(N + R_{N-1}) \le s$ with $R_{N-1} = d_{N-1}/(1-d_{N-1})$; this is the stated condition. In particular $tN \le s$. Since $c_i = (r+ti)d_i$ and $b_i = (s-ti)d_i$, the Leibniz rule for differences gives, for $\ell \ge 0$ and $1 \le n \le N - \ell$, which includes every index occurring in the moves on Ω,

$$\varphi_n(\ell) = (r + t(\ell+n))f_n(\ell) + tnf_{n-1}(\ell), \qquad \psi_n(\ell) = h_n(\ell) - v_n(\ell),$$

with $h_n(\ell) = (s - t(\ell+n))f_n(\ell)$ and $v_n(\ell) = tnf_{n-1}(\ell)$; both are nonnegative when $\ell + n \le N$. (Outside this range the formulas need not hold, because the Newton coefficients of b and c are set to zero beyond N.) Extend ν by zero outside $1 \le i \le N$ and put $\widehat{\kappa}_i = p_i\nu_i + q_i\nu_{i-1}$ $(i \ge 1)$, $\widehat{\kappa}_0 = 0$, with $p_i = r + ti$ and $q_i = ti$; the Newton sequence κ of c is the truncation of $\widehat{\kappa}$ at N. Then

$$\widehat{\kappa}_i^2 - \widehat{\kappa}_{i-1}\widehat{\kappa}_{i+1} = p_i^2(\nu_i^2 - \nu_{i-1}\nu_{i+1}) + q_i^2(\nu_{i-1}^2 - \nu_{i-2}\nu_i) + q_{i-1}p_{i+1}(\nu_i\nu_{i-1} - \nu_{i-2}\nu_{i+1}) + t^2(\nu_{i-1}\nu_{i+1} + \nu_{i-2}\nu_i + 2\nu_i\nu_i$$

and each term is nonnegative by log-concavity of ν. The positive support of $\widehat{\kappa}$ is that of ν when $t = 0$ and extends it by one index when $t > 0$, an interval in both cases. Truncation at N preserves log-concavity and interval support, so κ is log-concave without internal zeros, and Lemma 2.1 applies to f and to φ.

Moves. We use the recurrence of the proof of Theorem 3.1, which holds for every array. For $j \ge 1$ split $A_j^{\psi} = H_j - V_j$, where $H_j = W^h_{2j+1,j}$ and $V_j = W^v_{2j+1,j}$ are formed from h_n and v_n in place of ψ_n; both are nonnegative on Ω. Keep $A_0^{\psi}(\ell, a) = b_{\ell+a} - b_\ell$ as one move; it is nonnegative because b is nondecreasing. The weights B_j, formed from φ, and $C_j + C_j^{\psi} = W^f_{2j+2,j+1}$, formed from f, are nonnegative. Write $C'_j = C_j + C_j^{\psi}$.

Detours. All detour inequalities below are asserted for source nodes $(\ell, a) \in \Omega$; in the mixed products the occurring indices satisfy $n \le a - 2$, hence $\ell + n + 2 \le N$. Since $\varphi_2(\ell) \ge (r + t(\ell+1))f_2(\ell)$ and $c_{\ell+1} = (r + t(\ell+1))d_{\ell+1}$, Lemma 2.1 for f gives

$$\varphi_2(\ell)f_n(\ell+1) \ge c_{\ell+1}f_{n+2}(\ell), \quad \varphi_2(\ell)h_n(\ell+1) \ge c_{\ell+1}h_{n+2}(\ell), \quad \varphi_2(\ell)v_n(\ell+1) \ge \tfrac{n}{n+2}\,c_{\ell+1}v_{n+2}(\ell)\ (n \ge 2),$$

using $s - t(\ell+n+1) \ge s - t(\ell+n+2) \ge 0$ for the second and Lemma 2.1 with indices $(2, n-1)$ for the third. As in the proof of [1, Lemma 3.4], with $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell)$, for

$0 \le \ell \le N-1$ and second factors at $(\ell+1, a-2)$,

$$B_0 B_{j-1} \ge 4c_{\ell+1} B_j \ (j \ge 1), \quad B_0 H_{j-1} \ge 6c_{\ell+1} H_j \ (j \ge 2), \quad B_0 V_{j-1} \ge 6c_{\ell+1} V_j \ (j \ge 2), \quad B_0 C'_{j-1} \ge 6c_{\ell+1} C'_j \ (j$$

The first uses Lemma 2.1 for φ. The coefficient ratios are, on $m \ge 2j+2$, $2j+1$, $2j+1$ and $2j+2$ respectively,

$$\frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{jm(m-1)}{2(m-j-1)}, \qquad \frac{j(m-1)(m-2)}{2(m-j-1)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)};$$

they increase with j and are at least 5, 10, 6 and 6. Finally,

$$A_0^{\psi}(\ell,a) B_0(\ell+1,a-1) + B_0(\ell,a) A_0^{\psi}(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1(\ell,a) - 2V_1(\ell,a)\big). \quad (4.1)$$

Only $a \ge 3$ needs proof. Put $u_n = f_n(\ell)$, $y_n = \binom{a}{n} u_n$, $D = u_0 + u_1 = d_{\ell+1}$, $\alpha = s - t(\ell+a)$, $X = \sum_{n\ge1} y_n$, $Y = \sum_{n\ge2} y_n$, $X' = \sum_{j\ge1} \binom{a-2}{j}(u_j + u_{j+1})$, $Y' = \sum_{j\ge2} \binom{a-1}{j}(u_j + u_{j+1})$ and $J = \sum_{n=2}^{a-1} \binom{a-1}{n} u_n$, so that $X = au_1 + Y$, $X' = (a-2)u_1 + J$ and $Y' = Y - (a-1)u_2$. Then $A_0^{\psi}(\ell,a) = \alpha X - tau_0$ and $A_0^{\psi}(\ell+1,a-2) = (\alpha+t)X' - t(a-2)D$; both are nonnegative, and the two weights B_0 are at least $(r + t(\ell+1))Y'$ and $(r + t(\ell+1))Y$. With $H_* = \sum_{n\ge3}(n-2)y_n$ and $T = \sum_{n\ge2}(n-3)(a-n)y_n$ one has $H_1 = \sum_{n\ge3}(n-2)(\alpha + t(a-n))y_n$ and, after a shift of index, $V_1 = t\sum_{n\ge2}(n-1)(a-n)y_n$, so $4H_1 - 2V_1 = 4\alpha H_* + 2tT$. Since $c_{\ell+1} = (r + t(\ell+1))D$, it suffices to show

$$(\alpha X - tau_0)Y' + Y\big((\alpha+t)X' - t(a-2)D\big) \ge D(4\alpha H_* + 2tT). \quad (4.2)$$

The coefficient of α in the difference of the two sides is $K = XY' + YX' - 4DH_*$. By Lemma 2.1, $u_i(u_j + u_{j+1}) \ge Du_{i+j}$ for $i, j \ge 1$, so $XY' \ge au_1 Y' \ge D\sum_{n\ge3} n y_n$ and $YX' \ge \binom{a}{2} u_2 X' \ge D\sum_{n\ge3}\binom{n}{2} y_n$; hence $K \ge \frac{D}{2}\sum_{n\ge3}\big((n-3)(n-4)+4\big)y_n \ge 0$, and (4.2) holds when $t = 0$. Let $t > 0$. Monotonicity of b at $\ell + a - 1$ gives $\alpha \ge tR_{\ell+a-1} \ge tR_\ell = tu_0/u_1$, so it suffices to prove (4.2) at $\alpha = tu_0/u_1$. Both sides are homogeneous of degree two in (u_n); take $u_1 = 1$ and write $z = u_0$ and $q = u_2$. If $q = 0$, then $u_n = 0$ for $n \ge 2$ and both sides vanish. Otherwise $0 \le z \le 1/q$ by log-concavity. At $\alpha = tz$ the two weights A_0^{ψ} are tzY and $t(z+1)J$, and (4.2) divided by t reads $F(z) \ge 0$, where

$$F(z) = zYY' + (z+1)YJ - (z+1)(4zH_* + 2T) = (1-qz)F(0) + qzF(1/q) + 4H_* z(1/q - z),$$

the second form holding because F is quadratic with leading coefficient $-4H_*$. It remains to show $F(0) \ge 0$ and $F(1/q) \ge 0$. Put $L_3 = \frac{3}{2}$, $L_n = \frac{n(n+2)}{6}$ for $n \ge 4$, and $E_n = \frac{2(n^2-5)}{n+1}$. In YJ keep, for each n, the product with indices $(2,2)$ if $n = 3$, and the products with indices $(2, n-1)$ and $(3, n-2)$ if $n \ge 4$; by log-concavity with $u_1 = 1$, $u_i u_{n+1-i} \ge u_n$, and $\binom{a-1}{j} \ge \binom{a-i}{j}$, so $YJ \ge \sum_{n=3}^{a-1} L_n(a-n)y_n$. In Y keep the term $\binom{a}{2}q$; as $q(u_{n-1} + u_n) \ge (q+1)u_n$ for $n \ge 3$, $YY' \ge (q+1)\sum_{n=3}^{a} \frac{n(a-1)}{2} y_n$. Since $L_n \ge 2(n-3)$,

$$F(0) = YJ - 2T \ge 2(a-2)y_2 + \sum_{n=3}^{a-1}\big(L_n - 2(n-3)\big)(a-n)y_n \ge 0.$$

Using $u_n \le q u_{n-1}$ and a shift of index, $\frac{4}{q}H_* + 2T \le \sum_{n=2}^{a-1} E_n(a-n)y_n$, where $E_2 < 0$. As $L_n + \frac{n}{2} \ge E_n$ for $n \ge 3$,

$$\frac{F(1/q)}{1+1/q} = YJ + \frac{YY'}{q+1} - \frac{4}{q}H_* - 2T \ge \sum_{n=3}^{a-1}\Big(L_n + \frac{n}{2} - E_n\Big)(a-n)y_n \ge 0.$$

The numerical facts used are $L_3 = \frac{3}{2} \ge 0$, $L_3 + \frac{3}{2} = 3 \ge E_3 = 2$, and, for $n \ge 4$, $6(L_n - 2(n-3)) = (n-5)^2 + 11$ and $6(n+1)(L_n + \frac{n}{2} - E_n) = n^3 - 6n^2 + 5n + 60 > 0$.

Resummation. Regard the moves as operators on functions on Ω, each divided by c at the destination level; all strictly raise ℓ, and inequalities between operators are entrywise. Let P and Q be the sums of the B_j with j even and odd, $C = C_+ - C_-$ with $C_\pm$ the sums of the C'_j with j even and odd, H_+ and H_- the sums of the H_j with $j \ge 2$ even and $j \ge 3$ odd, V_+ and V_- the sums of the V_j with j odd and $j \ge 2$ even, A_0 the operator of A_0^ψ, and U the straight moves. The recurrence reads $\omega = \beta + (U + P - Q + C + A)\omega$ with $A = A_0 + H_+ - H_- - H_1 + V_+ - V_-$ and $\beta = \omega|_{\ell=0} \ge 0$. The inequalities above give

$$P^2 \ge 4Q, \qquad H_+P \ge 6H_-, \qquad V_+P \ge 6V_-, \qquad C_+P \ge 6C_-, \qquad A_0P + PA_0 + 2V_1 \ge 4H_1.$$

Let $G = (I - P/2)^{-1} = \sum_{m=0}^{N}(P/2)^m \ge 0$. Then $PG = GP = 2(G - I)$ and $(I - P/6)G = \frac{2}{3}G + \frac{1}{3}I$, so $G(H_+ - H_-)G \ge GH_+(I - P/6)G \ge 0$, $GCG \ge 0$ and $G(V_+ - V_-)G \ge \frac{2}{3}GV_1G$. Hence

$$GAG \ge G\big(A_0 - H_1 + \tfrac{2}{3}V_1\big)G \ge G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G = \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \ge 0.$$

With $R = U + P^2/4 - Q \ge 0$, the operator $M = G(R + C + A)G$ is nonnegative and strictly raises ℓ. Since $(I - P/2)^2\omega = \beta + (R + C + A)\omega$, the function $v = (I - P/2)\omega$ satisfies $v = G\beta + Mv$. Therefore $\omega = G\sum_{q=0}^{N} M^q G\beta \ge 0$, and $g(m,n;k) \ge 0$ as in the proof of Theorem 3.1. ∎

Theorem 4.2 *(Convex quadratic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $\rho(x) = r + tx + \tau\binom{x}{2}$ with $r > 0$ and $t, \tau \ge 0$, and put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 4.1 ($\tau = 0$). As in the proof of that theorem, $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - (t + \tau i)R_i\big)$ with a nonincreasing bracket, so b is nondecreasing exactly when $t(N + R) + \tau\big(\binom{N}{2} + (N-1)R\big) \le 1 - r$ with $R = d_{N-1}/(1 - d_{N-1})$. The Newton coefficients of c need not be log-concave: for $d_i = i/4$ and $\rho(x) = \frac{1}{2} + \binom{x}{2}/40$ they are $(1/8, 1/80, 3/160, 0)$. The proof below replaces their log-concavity by a weighted detour inequality.

Proof. Write $f_n(\ell) = \Delta^n d(\ell)$, $\Delta\rho(x) = t + \tau x$ and $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 4.1. Since $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - \Delta\rho(i)R_i\big)$ and b is nondecreasing, $1 - \rho(m+1) \ge \Delta\rho(m)R_m$ for $0 \le m < N$, and $\rho(i) \le 1$ for $i \le N$. For $\ell + n \le N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \binom{n}{2}\tau f_{n-2}, \qquad \psi_n = h_n - v_n - w_n,$$

with $h_n = (1-\rho(\ell+n))f_n$, $v_n = n\Delta\rho(\ell+n-1)f_{n-1}$ and $w_n = \binom{n}{2}\tau f_{n-2}$; here φ_n, h_n, v_n and w_n are nonnegative, while ψ_n need not be.

Moves. In the recurrence of the proof of Theorem 3.1, keep $A_0^\psi = b_{\ell+a} - b_\ell \ge 0$ and split $A_j^\psi = H_j - V_j - W_j$ for $j \ge 1$, where H_j, V_j and W_j are $W_{2j+1,j}$ formed from h_n, v_n and w_n; the weights B_j, formed from φ, and $C'_j = C_j + C_j^\psi = W^f_{2j+2,j+1}$ are nonnegative. With the signs $(-1)^j$, the moves W_j behave like the V_j.

Detours of H, V, W and C'. All detours are asserted for sources $(\ell,a) \in \Omega$, with second factors at $(\ell+1, a-2)$, marked by a prime. Since $\varphi_2(\ell) \ge \rho(\ell+1)f_2(\ell)$ and $c_{\ell+1} = \rho(\ell+1)d_{\ell+1}$, Lemma 2.1 for f gives, for $n \le a-2$,

$$\varphi_2 f'_n \ge c_{\ell+1} f_{n+2}, \quad \varphi_2 h'_n \ge c_{\ell+1} h_{n+2}, \quad (n+1)(n+2)\varphi_2 v'_n \ge n^2 c_{\ell+1} v_{n+2}, \quad (n+1)(n+2)\varphi_2 w'_n \ge n(n-1)c_{\ell+1}$$

the last two for $n \ge 3$. The second uses $1-\rho(\ell+n+1) \ge 1-\rho(\ell+n+2) \ge 0$; the third uses Lemma 2.1 with indices $(2, n-1)$ and $(n+1)\Delta\rho(\ell+n) - n\Delta\rho(\ell+n+1) = t + \tau\ell \ge 0$; the fourth uses indices $(2, n-2)$. As in the proof of [1, Lemma 3.4], with $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$ and $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$,

$$B_0 H'_{j-1} \ge 10 c_{\ell+1} H_j, \qquad B_0 V'_{j-1} \ge \tfrac{9}{2} c_{\ell+1} V_j, \qquad B_0 W'_{j-1} \ge 3 c_{\ell+1} W_j \quad (j \ge 2), \qquad B_0\, C'_{j-1}(\ell+1, a-2) \ge 6c_{\ell+1}$$

The coefficient ratios are $\frac{jm(m-1)}{2(m-j-1)}$, $\frac{j(m-2)^2}{2(m-j-1)}$ and $\frac{j(m-2)(m-3)}{2(m-j-1)}$ on $m \ge 2j+1$, and $\frac{(j+1)m(m-1)}{2(m-j-1)}$ on $m \ge 2j+2$; they increase with j, and at the smallest j the bounds are $(m-5)(m-6) \ge 0$, $(m-5)(2m-7) \ge 0$, $m \ge 5$ and $(m-3)(m-4) \ge 0$.

Detour of B. We show, for $n \ge 2$ and $\ell + n + 2 \le N$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \ge 8(n-1)\,c_{\ell+1}\varphi_{n+2}(\ell). \tag{4.3}$$

Granting this, $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$, (4.3) with $n = m-2$, $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$ and $\binom{m}{2}\frac{8(m-3)}{m(m-1)} = 4(m-3)$ give

$$B_0(\ell,a)B_{j-1}(\ell+1,a-2) \ge 4c_{\ell+1}\sum_{m\ge 2j+2}(m-3)\binom{m-j-3}{j-1}\binom{a}{m}\varphi_m(\ell) \ge 4c_{\ell+1}B_j(\ell,a)$$

for $j \ge 1$, because $(m-3)\binom{m-j-3}{j-1} - \binom{m-j-2}{j} = \binom{m-j-3}{j-1}\frac{(j-1)(m-2)}{j} \ge 0$. (With constant 1 in place of $\frac{(n+1)(n+2)}{8(n-1)}$, (4.3) can fail: for $N = 20$, $d_i = (2^i-1)/(2^{20}-1)$, $r = \frac12$, $t = 0$, $\tau = 10^{-3}$ one gets $\varphi_2(16)\varphi_2(17) < c_{17}\varphi_4(16)$.)

To prove (4.3), put $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$, and $p = \rho(\ell+1)$, $K_n = (n+1)(n+2)$, $L_n = 8(n-1)$. Both sides are homogeneous of degree two in u; take $u_1 = 1$ and write $z = u_0$, $q = u_2$. By convexity and log-concavity, $0 \le z \le \ell$ and $zq \le 1$. If $q = 0$ or $u_n = 0$, then $\varphi_{n+2}(\ell) = 0$. Otherwise put $y = u_{n+1}/u_n \in [0,q]$. Log-concavity gives $u_{n+2} \le y^2 u_n$, and for $n \ge 3$ also $u_{n-1} \ge u_n/q$ and $u_{n-2} \ge u_n/q^2$. Hence $\varphi_n(\ell+1) \ge u_n\mathcal{L}(y)$ and $\varphi_{n+2}(\ell) \le u_n\mathcal{T}(y)$, where

$$\mathcal{T}(y) = \rho(\ell+n+2)y^2 + (n+2)\Delta\rho(\ell+n+1)y + \binom{n+2}{2}\tau,$$

$\mathcal{L}(y) = \rho(\ell+n+1)(1+y) + n\Delta\rho(\ell+n)(1+q^{-1}) + \binom{n}{2}\tau(q^{-1}+q^{-2})$ for $n \ge 3$, and $\mathcal{L}(y) = \rho(\ell+3)(1+y) + 2\Delta\rho(\ell+2)(1+q^{-1}) + \tau(z+1)q^{-1}$ for $n = 2$ (exact). In both cases $\mathcal{L}(y) = A + By$ with $A \ge B \ge 0$, so $\mathcal{L}(0) \ge \mathcal{L}(q)/(1+q)$; and $\mathcal{T}(q) \ge (1+q)\mathcal{T}(0)$ because $(n+2)\Delta\rho(\ell+n+1) \ge (n+2)(n+1)\tau$. The function $\mathcal{F}(y) = K_n\varphi_2(\ell)\mathcal{L}(y) - L_n p(z+1)\mathcal{T}(y)$ is concave with $\mathcal{F}(0) \ge \mathcal{F}(q)/(1+q)$, so it suffices that $\mathcal{F}(q) \ge 0$, which is (4.3) for the configuration $u_0 = z$, $u_k = q^{k-1}$ $(k \ge 1)$.

Next we reduce to $z = \min(\ell, 1/q)$. For $n \ge 3$, only $\varphi_2(\ell) = A_2 + \tau z$ and $c_{\ell+1} = p(z+1)$ depend on z, where $A_2 = \rho(\ell+2)q + 2\Delta\rho(\ell+1) \ge \tau$, so $\varphi_2(\ell)/(z+1)$ is nonincreasing in z. For $n = 2$, write also $\varphi_2(\ell+1) = G_2 + \tau(z+1)$ with $G_2 = \rho(\ell+3)q(1+q) + 2\Delta\rho(\ell+2)(1+q)$. For $z \le Z \le \ell$ the quotient $\varphi_2(\ell)\varphi_2(\ell+1)/(z+1)$ at z minus its value at Z equals $(Z-z)\big(G_2(A_2-\tau)/((z+1)(Z+1)) - \tau^2\big) \ge 0$, since $G_2(A_2-\tau) \ge 2(\ell+2)(2\ell+1)\tau^2 \ge (\ell+1)^2\tau^2$.

For the remaining configurations we use the following polynomial inequality. For an integer $n \ge 2$ and reals $h \ge 1$, $p, \tau \ge 0$ and $s \ge (h-1)\tau$, put $A = p + (2h-1)s + h^2\tau$, $B = p + nh(s+\tau) + \binom{n}{2}h^2\tau$ and $C = p + ((n+2)h-1)s + \binom{n+2}{2}h^2\tau$. Then

$$K_n AB - L_n pC \ge 0. \tag{4.4}$$

Indeed, $K_nAB - L_npC$ is nondecreasing in s (its ps-coefficient is $a_n((n+2)h-1)$ with $a_n = n^2 - 5n + 10 > 0$, and its other s-dependent coefficients are nonnegative), and at $s = (h-1)\tau$ it equals $a_np^2 + C_n(h)p\tau + D_n(h)\tau^2$, where $C_n(h) = \frac{1}{2}\big(h^2U_n + h(10n^2+14n-60) + 2a_n\big)$, $U_n = n^4 - 4n^3 - 21n^2 + 12n + 60$ and $D_n(h) = \frac{1}{2}h^2n(n+1)^2(n+2)(3h^2-3h+1)$. For $n \ge 7$ all three coefficients are nonnegative, since $U_n = k^4 + 24k^3 + 189k^2 + 502k + 144$ with $k = n-7$. For $2 \le n \le 6$, $4a_nD_n - C_n^2$ is a polynomial in $v = h-1$ with positive coefficients:

n	v^0	v^1	v^2	v^3	v^4
2	576	2880	5616	4992	1664
3	1520	6240	9744	7200	2160
4	6300	27000	42660	30600	8604
5	25200	126000	229500	184800	56000
6	62720	376320	774144	677376	216576

so $a_np^2 + C_np\tau + D_n\tau^2 = a_n\big(p + \frac{C_n\tau}{2a_n}\big)^2 + \frac{4a_nD_n - C_n^2}{4a_n}\tau^2 \ge 0$.

If $1/q \le \ell$, take $z = 1/q$, $h = 1 + 1/q$ and $s = t + \tau\ell \ge (h-1)\tau$. For the configuration above, $\mathcal{F}(q)$ divided by $q(1+q)$ equals $K_nAB - L_npC$, which is nonnegative by (4.4).

If $\ell < 1/q$, take $z = \ell$, and write $w = 1/q - \ell > 0$, $h_0 = \ell + 1$, $h = h_0 + w$ and $s = t + \tau\ell$. With $A(w) = p + (2h-1)s + \tau\big((\ell+2)h - \ell - 1\big)$ and $B(w)$, $C(w)$ as above (with this h), $\mathcal{F}(q)$ divided by q^2 equals $\mathcal{E}_n(w) = K_n(h_0+w)A(w)B(w) - L_nh_0pC(w)$ for $n \ge 3$, and $\mathcal{E}_2(w) - 12A(w)\tau w(\ell+w)^2$ for $n = 2$. At $w = 0$ both equal $h_0(K_nAB - L_npC)$ with $h = h_0$, which is nonnegative by (4.4). Expanding in w, with $\bar{A} = A(0) - p$ and $\bar{B} = B(0) - p$, every coefficient of w, w^2, w^3 and w^4 is a sum of nonnegative terms: for $n \ge 3$,

$$[w] = K_np^2 + p(a_1t + b_1\tau) + K_n\big(\bar{A}\bar{B} + h_0(\eta\bar{B} + \bar{A}\theta)\big),$$
$$[w^2] = p(a_2t + b_2\tau) + K_n\big(\eta\bar{B} + \bar{A}\theta + h_0(\eta\theta + \bar{A}\epsilon)\big), \qquad [w^3] = K_n\big(\eta\theta + A(0)\epsilon + h_0\eta\epsilon\big), \qquad [w^4] = K_n\eta\epsilon,$$

where $\eta = 2s + (\ell+2)\tau$, $\theta = n(s+\tau) + n(n-1)\tau h_0$, $\epsilon = \binom{n}{2}\tau$, $a_1 = (n+2)\big(2\ell(n^2 - n + 6) + 2n^2 - 3n + 11\big)$, $a_2 = K_n(n+2)$, $b_1 = \frac{n+2}{2}(\ell^2 P_n + \ell Q_n + R_n)$ and $b_2 = \frac{K_n}{2}\big(3n(n-3)(\ell+1) + 14\ell + 12\big)$, with $P_n = 3k^3 + 15k^2 + 6k + 8$, $Q_n = 6k^3 + 30k^2 + 20k + 16$ and $R_n = 3k^3 + 15k^2 + 16k + 16$ for $k = n-3$; for $n = 2$ the coefficients of w, w^2 and w^3 are

$$12p^2 + 4(16\ell+13)pt + 8(\ell+1)(2\ell+3)p\tau + 48(\ell+1)(3\ell+2)t^2 + 12(36\ell^3 + 92\ell^2 + 76\ell + 21)t\tau + 12(27\ell^4 + 93\ell^3 + 114\ell^2 + 62\ell + 13)\tau^2,$$

$$48pt + 12(2\ell+3)p\tau + 24(6\ell+5)t^2 + 12(36\ell^2 + 67\ell + 29)t\tau + 12(27\ell^3 + 78\ell^2 + 67\ell + 19)\tau^2,$$

$$12\big(2s + (\ell+2)\tau\big)\big(2t + (3\ell+5)\tau\big).$$

This proves (4.3).

Key inequality. We show, for $(\ell, a) \in \Omega$ with $a \geq 3$,

$$A_0^\psi(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^\psi(\ell+1,a-2) \geq c_{\ell+1}\big(4H_1 - 2V_1 - W_1\big)(\ell,a). \quad (4.5)$$

Use the notation of the proof of Theorem 4.1 ($u_n = f_n(\ell)$, y_n, D, X, Y, X', Y', J, H_*, T), and put $\alpha = 1 - \rho(\ell+a)$, $s = \Delta\rho(\ell+a-1)$, $p = \rho(\ell+1)$ and $Z = \sum_{n\geq 2}\binom{a-n}{2}y_n$. Then $\alpha \geq sR_{\ell+a-1} \geq su_0/u_1$ and $s \geq \tau(a-1)$. Backward differences give

$$A_0^\psi(\ell,a) = \alpha X - sau_0 + \tau\tbinom{a}{2}u_0, \qquad A_0^\psi(\ell+1,a-2) = (\alpha+s)X' - s(a-2)D + \tau\tbinom{a-1}{2}D,$$

both nonnegative, and $B_0(\ell,a) \geq pY$, $B_0(\ell+1,a-1) \geq pY'$, $c_{\ell+1} = pD$. Since $1 - \rho(\ell+n) = \alpha + (a-n)s - \tau\binom{a-n}{2}$, reindexing gives $4H_1 - 2V_1 = 4\alpha H_* + 2sT + 4\tau Z$ and $W_1 = \tau\sum_{n\geq 1} n\binom{a-n}{2}y_n$. The argument of the proof of Theorem 4.1 with s in place of t, keeping the surplus of the two endpoints ($F(0) \geq 2(a-2)y_2$ and $F(1/q) \geq \frac{2}{3}(1+1/q)(a-2)y_2$, from the term $n = 2$ and $E_2 = -\frac{2}{3}$), shows

$$(\alpha X - sau_0)Y' + Y\big((\alpha+s)X' - s(a-2)D\big) - D(4\alpha H_* + 2sT) \geq \tfrac{2}{3}sD(a-2)y_2$$

for $\alpha u_1 \geq su_0$; the right side of this bound is linear in $z = u_0/u_1$ and is subtracted before the interpolation. With $s \geq \tau(a-1)$, the difference of the two sides of (4.5) divided by p is therefore at least

$$\tau\Big(\tbinom{a}{2}u_0Y' + D\Big[\tbinom{a-1}{2}y_1 + \tfrac{(a-2)(a+11)}{6}y_2 + (2a-5)y_3 + \sum_{n\geq 4}\Big(\tbinom{a-1}{2} + (n-4)\tbinom{a-n}{2}\Big)y_n\Big]\Big) \geq 0.$$

Resummation. With $W_\pm$ the sums of the W_j with j odd and $j \geq 2$ even, the recurrence reads as in the proof of Theorem 4.1 with $A = A_0 - H_1 + (H_+ - H_-) + (V_+ - V_-) + (W_+ - W_-)$, and the inequalities above give $P^2 \geq 4Q$, $H_+P \geq 10H_-$, $V_+P \geq \frac{9}{2}V_-$, $W_+P \geq 3W_-$, $C_+P \geq 6C_-$ and $A_0P + PA_0 + 2V_1 + W_1 \geq 4H_1$. Since $(I - P/\mu)G = (1 - \frac{2}{\mu})G + \frac{2}{\mu}I$, one has $G(H_+ - H_-)G \geq 0$, $GCG \geq 0$, $G(V_+ - V_-)G \geq \frac{5}{9}GV_1G$ and $G(W_+ - W_-)G \geq \frac{1}{3}GW_1G$, hence

$$GAG \geq G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{18}V_1 + \tfrac{1}{12}W_1\big)G = \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{18}GV_1G + \tfrac{1}{12}GW_1G \geq 0.$$

With $U + P^2/4 - Q \geq 0$ this gives $M \geq 0$, and (P) follows as before. ∎

5. Ratios of higher degree

For profiles of Newton degree at most two the ratio can have any degree, provided the Newton coefficients of c are nonnegative and log-concave without internal zeros; no key inequality is needed.

Theorem 5.1 *(Low-degree profiles).* *Let $d_i = \nu_1 i + \nu_2 \binom{i}{2}$ with $\nu_1 > 0$, $\nu_2 \geq 0$ and $d_N = 1$, let $c_0 = 0$, and suppose that the Newton coefficients $\kappa_k = \Delta^k c(0)$, $1 \leq k \leq N$, are log-concave without internal zeros. Put $a_i = 1 - d_i$ and $b_i = d_i - c_i$. If b is nondecreasing, then the system has nonnegative linearization.*

Proof. We use the recurrence of the proof of Theorem 3.1, which holds for every array, with $\varphi_n(\ell) = \Delta^n c(\ell)$, $\psi_n(\ell) = \Delta^n b(\ell)$ and $f_n(\ell) = \Delta^n d(\ell)$ for the polynomial extensions. Since $f_n = 0$ for $n \geq 3$, every move A_j^{ψ} with $j \geq 1$, which involves only $n \geq 2j+1 \geq 3$, equals $-\widetilde{B}_j$ with $\widetilde{B}_j = W_{2j+1,j}$ formed from φ; the moves $C_j + C_j^{\psi} = W_{2j+2,j+1}^{f}$ vanish for $j \geq 1$, and $C_0 + C_0^{\psi} = \binom{a}{2}\nu_2 \geq 0$. The weight $A_0^{\psi}(\ell, a) = b_{\ell+a} - b_\ell$ is nonnegative on Ω, as are the straight weights and all B_j and $\widetilde{B}_j$, since $(\varphi_n(\ell))_n$ is obtained from (κ_k) by the shifts of Lemma 2.1 and is therefore nonnegative and log-concave without internal zeros. Lemma 2.1 for φ gives, as in [1, Lemma 3.4], $B_0(\ell, a)B_{j-1}(\ell+1, a-2) \geq 4c_{\ell+1}B_j(\ell, a)$ for $j \geq 1$, and with $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell)$ also $B_0(\ell, a)\widetilde{B}_{j-1}(\ell+1, a-2) \geq 4c_{\ell+1}\widetilde{B}_j(\ell, a)$ for $j \geq 2$, because the coefficient quotients $jm(m-1)/(2(m-j-1))$ on $m \geq 2j+1$ are at least 10. In the resummation of the proof of Theorem 4.1 take $H = 0$ and $V_j = \widetilde{B}_j$; the key inequality is then trivial, and $M \geq 0$ gives (P). ∎

Corollary 5.2. *In Theorem 5.1 let $c_i = \rho(i)d_i$ with $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k\geq 0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Then the Newton coefficients of c are log-concave without internal zeros. Hence every such ratio, of any degree, with nondecreasing b gives nonnegative linearization on these profiles.*

Proof. Put $s_k = \rho_k + \rho_{k-1}$ ($\rho_{-1} = 0$), which is log-concave without internal zeros, and $h = \nu_2/2$. The Leibniz rule gives $\kappa_k = k e_k$ for $k \geq 1$, with $e_k = (\nu_1 + h(k-1))s_k + h(k-1)s_{k-1}$. Write $\alpha = \nu_1 + h(k-1)$ and $\beta = h(k-1)$; then

$$\begin{aligned} e_k^2 - e_{k-1}e_{k+1} = {} & \alpha^2(s_k^2 - s_{k-1}s_{k+1}) + \beta^2(s_{k-1}^2 - s_{k-2}s_k) + (\beta - h)(\alpha + h)(s_{k-1}s_k - s_{k-2}s_{k+1}) \\ & + h^2(s_{k-1}s_{k+1} + s_{k-2}s_k + 2s_{k-1}s_k), \end{aligned}$$

and every term is nonnegative for $k \geq 2$ by log-concavity of s. So $\kappa_k^2 \geq k^2 e_{k-1}e_{k+1} \geq \kappa_{k-1}\kappa_{k+1}$. If $\kappa_k = 0$ with $k \geq 1$, then $s_k = 0$, so $\rho_k = 0$; as $\rho_0 > 0$ and ρ has no internal zeros, $\rho_m = 0$ for $m \geq k$, and $\kappa_{k+1} = 0$. ∎

For example, on the linear profile $d_i = i/N$ the exponential ratios $\rho(x) = r(1+\varepsilon)^x$ with $r > 0$ and $\varepsilon \geq 0$, whose Newton coefficients $r\varepsilon^k$ are geometric, give nonnegative linearization whenever b is nondecreasing. Nonnegativity of the ratio's Newton coefficients alone does not suffice: the cubic counterexample in Section 6 has the linear profile $d_i = i/4$ and the ratio with Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$, which are nonnegative and satisfy the

log-concavity inequalities but have internal zeros. Theorem 4.2 and Corollary 5.2 are not comparable: the former admits convex quadratic ratios with $t^2 < r\tau$ on every log-concave Newton profile, the latter ratios of every degree on profiles of Newton degree at most two.

Products of Newton sequences. The proof of Corollary 5.2 used only that the Newton coefficients of $c = \rho d$ are log-concave. This holds for every profile.

Proposition 5.3 *(Products).* *Let $(\rho_i)_{i\geq 0}$ and $(\nu_j)_{j\geq 0}$ be nonnegative and log-concave without internal zeros, extended by zero at negative indices, and let $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$ for $x \in \mathbb{Z}_{\geq 0}$, where the sums are finite. Then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of fg are nonnegative and log-concave without internal zeros.*

Proof. The product rule $\binom{x}{i}\binom{x}{j} = \sum_k \binom{k}{i}\binom{i}{k-j}\binom{x}{k}$ gives $\kappa_k = \sum_p H_k(p)\rho_{p_1}\nu_{p_2}$, where $H_k(i,j) = \binom{k}{i}\binom{i}{k-j} = [y^i w^j](y + w + yw)^k$ vanishes unless $i, j \leq k \leq i + j$. If either sequence vanishes identically, so does κ. Otherwise, if ρ and ν are supported on $[p_0, p_1]$ and $[q_0, q_1]$, with $p_1, q_1 \leq \infty$, then $\kappa_k > 0$ exactly for $\max(p_0, q_0) \leq k \leq p_1 + q_1$, so κ has no internal zeros. For log-concavity fix $k \geq 1$; then

$$\kappa_k^2 - \kappa_{k-1}\kappa_{k+1} = \sum_{c\in\mathbb{Z}^2}\sum_{p\in\mathbb{Z}^2} T_c(p)\, u_c(p_1)\, w_c(p_2), \qquad T_c(p) = H_k(p)H_k(c-p) - H_{k-1}(p)H_{k+1}(c-p),$$

with $u_c(i) = \rho_i\rho_{c_1-i}$ and $w_c(j) = \nu_j\nu_{c_2-j}$. By log-concavity without internal zeros, u_c is symmetric about $c_1/2$ and nondecreasing on $i \leq c_1/2$, so $u_c = \sum_{0\leq a\leq c_1/2}(u_c(a) - u_c(a-1))\mathbf{1}_{[a,c_1-a]}$ with nonnegative coefficients, and likewise w_c. It therefore suffices to show $D_c(B) = \sum_{p\in B} T_c(p) \geq 0$ for every box $B = [a, c_1 - a] \times [b, c_2 - b]$ with $a \leq c_1/2$, $b \leq c_2/2$.

Let $E = \{(1,0),(0,1),(1,1)\}$, so that $H_k = \sum_{e\in E} H_{k-1}(\cdot - e)$. Expanding $H_k(p)$, $H_k(c-p)$ and $H_{k+1}(c-p)$ in this way gives

$$D_c(B) = \sum_{e,e'\in E}\big(V_{e+e'}(B-e) - V_{e+e'}(B)\big), \qquad V_s(p) = H_{k-1}(p)H_{k-1}(c-s-p),$$

where $V_s(X) = \sum_{p\in X} V_s(p)$. Fix $s = e + e'$, write $V = V_s$, $m = c - s$, $t = k - 1$ and $\mu = m/2$. With $n_1 = 2t - m_2$, $n_2 = 2t - m_1$, $n_3 = m_1 + m_2 - 2t$ one has $V(x,y) = \frac{t!^2}{n_1!n_2!n_3!}\binom{n_1}{t-y}\binom{n_2}{t-x}\binom{n_3}{x+y-t}$ (and $V = 0$ if some $n_i < 0$). Three properties follow.

(T1) For every interval W, the marginal $G_W(x) = \sum_{y\in W} V(x,y)$ is a constant times $\binom{n_2}{t-x}\sum_{z\in t-W}\binom{n_1}{z}\binom{n_3}{x-z}$, a product of a binomial row and a convolution of two log-concave sequences without internal zeros; so G_W is log-concave without internal zeros. The same holds with the roles of the coordinates exchanged.

(T2) $V(c-s-p) = V(p)$, and V is reverse regular of order two: $V(x,y)V(x',y') \leq V(x,y')V(x',y)$ for $x \leq x'$, $y \leq y'$, by log-concavity of $\binom{n_3}{\cdot}$. Taking $x' = 2\mu_1 - x$ and y, y' symmetric about μ_2, central symmetry turns this into $V(x,y)^2 \leq V(x,y')^2$. Hence $V(x,y') \geq V(x,y)$ whenever $x \leq \mu_1$, $y \leq y'$ and $y + y' = 2\mu_2$; and symmetrically in the coordinates.

(T3) If G is log-concave without internal zeros and $G(x) \le G(2\mu_1 - x)$ for all $x \le \mu_1$, then G is nondecreasing on $x \le \mu_1$: otherwise $G(x-1) > G(x)$ for some $x \le \mu_1$, and log-concavity makes G nonincreasing from $x-1$ on, so $G(2\mu_1 - x + 1) \le G(x) < G(x-1)$.

Write $I = [a, c_1 - a]$, $J = [b, c_2 - b]$, $X(I', J') = V(I' \times J')$, and telescope along rows or columns. By the symmetry $H_k(i,j) = H_k(j,i)$ it suffices to treat five ordered pairs.

$e = e' = (1,0)$: $\mu = c/2 - (1,0)$ and $V(B-e) - V(B) = G_J(a-1) - G_J(a-2)$ by central symmetry. Since J is symmetric about μ_2, G_J is symmetric about μ_1, hence nondecreasing on $x \le \mu_1$ by (T1) and (T3); and $a - 1 \le \mu_1$.

$e = (1,0)$, $e' = (0,1)$: $\mu = (c - (1,1))/2$ and $V(B-e) - V(B) = V(a-1, c_2 - b) - V(a-1, b-1) \ge 0$ by (T2), since $a - 1 < \mu_1$ and $(c_2 - b) + (b-1) = 2\mu_2$.

$e = (1,0)$, $e' = (1,1)$: $\mu = c/2 - (1, \frac{1}{2})$ and

$$V(B-e) - V(B) = \big[V(a-1, c_2 - b) - V(a-1, b-1)\big] + \big[G_{J-1}(a-1) - G_{J-1}(a-2)\big].$$

The first bracket is nonnegative by (T2), as $a - 1 \le \mu_1$. For the second, $G_{J-1}(2\mu_1 - x) - G_{J-1}(x) = V(x, c_2 - b) - V(x, b-1) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) apply.

$e = (1,1)$, $e' = (1,0)$: the same V, and $V(B-e) - V(B) = \big[X(I-1, J-1) - X(I-1, J)\big] + \big[V(B - (1,0)) - V(B)\big]$. The first bracket vanishes by central symmetry, since $I - 1$ is symmetric about μ_1; the second is the previous case.

$e = e' = (1,1)$: $\mu = c/2 - (1,1)$ and $V(B-e) - V(B) = \big[K(b-1) - K(b-2)\big] + \big[X(a-1, J) - X(a-1, J-2)\big] + \big[G_{J-2}(a-1) - G_{J-2}(a-2)\big]$ with $K(y) = \sum_{x \in I-1} V(x,y)$. Since $I - 1$ is symmetric about μ_1, K is symmetric about μ_2 and nondecreasing on $y \le \mu_2$ by (T1) and (T3); and $b - 1 \le \mu_2$. The second bracket is $[V(a-1, c_2 - b) - V(a-1, b-2)] + [V(a-1, c_2 - b - 1) - V(a-1, b-1)]$, two differences of the form in (T2), since both pairs of second coordinates sum to $2\mu_2 = c_2 - 2$. For the third, $G_{J-2}(2\mu_1 - x) - G_{J-2}(x) = X(x, J) - X(x, J-2) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) give $G_{J-2}(a-1) \ge G_{J-2}(a-2)$.

Each of the nine ordered-step contributions $V_{e+e'}(B-e) - V_{e+e'}(B)$ is nonnegative, so $D_c(B) \ge 0$, which proves the proposition. (The individual $T_c(p)$ need not be nonnegative.) ∎

In the language of random sets: if S and T are independent random subsets of $[n]$ whose laws are invariant under permutations and whose cardinalities are ultra-log-concave of order n, then so is $|S \cup T|$, since $\Pr(|S \cup T| = k) = \binom{n}{k}\kappa_k$ for $\Pr(S = A) = \rho_{|A|}$ and $\Pr(T = A) = \nu_{|A|}$. This closure theorem is Walkup's Theorem 4 for two sets [4]: his random sets are conditionally uniform, and their fundamental sequences are ρ and ν. He derives it from his Theorem 2, which proves that the double-binomial convolution $h_k = \sum_i \sum_r \binom{k}{i}\binom{i}{r}\rho_i \nu_{k-r}$, which is κ_k after the substitution $j = k - r$, preserves log-concavity; Proposition 5.3 expresses it in the Newton basis, and the proof above is self-contained. Hoggar's theorem is the analogue for the power basis. The diagnostics `runs/np/two_step.py` and `runs/np/case_proofs.py` check in exact arithmetic the identity for $D_c(B)$, the sign of each of its nine terms for $k \le 9$, and every bracket of the case analysis, (T1) and (T2) for $k \le 6$.

Log-concave ratios on log-concave profiles. Corollary 5.2 extends to every log-concave Newton profile. Besides Proposition 5.3, the proof uses a sign criterion for quadratic forms in log-concave sequences.

Lemma 5.4. *Let $Q(u) = \sum_{0\le i\le j} q_{ij}u_iu_j$ with $q_{0j} \le 0$ for all j and $q_{ij} \ge 0$ for $2 \le i \le j$, and put $N_W(Q) = \sum_{i\le j,\, i+j=W} q_{ij}$. If $N_W(Q) \ge 0$ for every W, then $Q(u) \ge 0$ for every nonnegative sequence $(u_n)_{n\ge0}$ that is log-concave without internal zeros.*

Proof. Fix W and put $p_i = u_iu_{W-i}$ for $0 \le i \le W/2$ and $p_{-1} = 0$. Log-concavity without internal zeros makes p nondecreasing. With $T_k = \sum_{k\le i\le W/2} q_{i,W-i}$ we get $\sum_{i\le W/2} q_{i,W-i}p_i = \sum_{k\le W/2} T_k(p_k - p_{k-1})$. Here $T_0 = N_W(Q) \ge 0$, $T_1 = N_W(Q) - q_{0W} \ge 0$, and $T_k \ge 0$ for $k \ge 2$. ∎

The weight sums are the coefficients of Q on geometric sequences: $Q(1, r, r^2, \dots) = \sum_W N_W(Q)r^W$. This Abel argument, with the reduction to the full sum when the negative terms come first, is that of Wang and Yeh [5, Lemma 2.1 and Corollary 2.6].

Theorem 5.5 *(Log-concave ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k\ge0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 4.1 and Corollary 5.2. It also contains the case $t^2 \ge r\tau$ of Theorem 4.2. The exponential ratios $\rho(x) = r(1 + \varepsilon)^x$ are admitted on every log-concave Newton profile. The proof is computer-assisted in two places, both by exact rational arithmetic:

- the signs of finitely many explicit polynomials in the weight-sum certificates of step (vi);
- the key inequality (5.2) for windows $a \le 5$, where step (vii) uses exact linear-programming certificates.

Proof. Setup. Only $\rho_0, \dots, \rho_N$ enter, and truncating (ρ_k) after index N keeps it log-concave without internal zeros, so we may assume it finitely supported. Write $f_n(\ell) = \Delta^n d(\ell)$, $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 4.1, and $P_k(x) = \Delta^k\rho(x) = \sum_h \binom{x}{h}\rho_{k+h}$. As in the proof of Theorem 4.2, $1 - \rho(m+1) \ge \Delta\rho(m)R_m$ for $0 \le m < N$, and $0 < \rho_0 \le \rho(i) \le 1$ for $i \le N$; ρ is nondecreasing.

For every integer $x \ge 0$, the sequence $(P_k(x))_k$ arises from (ρ_k) by the shifts of Lemma 2.1, so it is log-concave without internal zeros. Since $\rho(x+1)^2 - \rho(x)\rho(x+2) = P_1(x)^2 - P_0(x)P_2(x) \ge 0$, the positive sequence $(\rho(n))_{n\ge0}$ is log-concave. (The function $x \mapsto \rho(x)$ of a real variable need not be: $\rho(x) = 1 + x/2 + x^2/2$ has $(\log\rho)''(0) > 0$.) Log-concavity in k gives $\rho(y)P_{k+1}(y) \le P_1(y)P_k(y)$. Combining these facts, for integers $0 \le x \le y$ and $k \ge 1$,

$$\rho(x)P_{k-1}(y+1) \le \rho(x+1)P_{k-1}(y), \qquad \rho(x)P_k(y) \le \Delta\rho(x)P_{k-1}(y). \tag{5.1}$$

Indeed, log-concavity of $x \mapsto \rho(x)$ gives $\rho(x)\rho(y+1) \le \rho(x+1)\rho(y)$, equivalently $\rho(x)P_1(y) \le \Delta\rho(x)\rho(y)$. For the first inequality, $\rho(y)P_{k-1}(y+1) = \rho(y)P_{k-1}(y) + \rho(y)P_k(y) \le \rho(y+1)P_{k-1}(y)$; multiply by $\rho(x)$ and divide by $\rho(y) > 0$. For the second, $\rho(x)\rho(y)P_k(y) \le \rho(x)P_1(y)P_{k-1}(y) \le \Delta\rho(x)\rho(y)P_{k-1}(y)$. By Proposition 5.3, the Newton coefficients of $c = \rho d$ are log-concave without internal zeros, and so is $(\varphi_n(\ell))_n$ for every ℓ. For $\ell + n \le N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \sum_{j=0}^{n} \binom{n}{j} P_j(\ell+n-j) f_{n-j}, \qquad \psi_n = h_n - v_n - \widehat{w}_n,$$

with $h_n = (1-\rho(\ell+n))f_n$, $v_n = n\Delta\rho(\ell+n-1)f_{n-1}$ and $\widehat{w}_n = \sum_{j=2}^{n} w_n^{(j)}$, where $w_n^{(j)} = \binom{n}{j} P_j(\ell+n-j) f_{n-j}$. Here φ_n, h_n, v_n, $w_n^{(j)}$ and $\widehat{w}_n$ are nonnegative, while ψ_n need not be.

Moves and detours. Split the moves as in the proof of Theorem 4.2, with W_j now formed from $\widehat{w}_n$. Lemma 2.1 for φ gives $B_0(\ell,a)B_{j-1}(\ell+1,a-2) \ge 4c_{\ell+1}B_j(\ell,a)$ for $j \ge 1$, as in [1, Lemma 3.4]. The detours of H and C', with constants 10 and 6, hold verbatim: they used only $\varphi_2(\ell) \ge \rho(\ell+1)f_2(\ell)$, Lemma 2.1 for f, and $0 \le \rho \le 1$ nondecreasing.

For V, the term $\rho(\ell+2)f_2$ of φ_2, the first inequality of (5.1) with $k=2$, $x = \ell+1$, $y = \ell+n$, and Lemma 2.1 for f with indices $(2, n-1)$ give

$$\varphi_2(\ell)v_n(\ell+1) \ge \tfrac{n}{n+2}\, c_{\ell+1}v_{n+2}(\ell).$$

For $\widehat{w}$, expand $\varphi_2(\ell) = \rho(\ell+2)f_2 + 2\Delta\rho(\ell+1)f_1 + P_2(\ell)f_0$, and pair its three terms with the components of $\widehat{w}_n(\ell+1)$:

- $\rho(\ell+2)f_2$ with component $i \le n$, by the first inequality of (5.1) and indices $(2, n-i)$, gives at least $\binom{n}{i}\binom{n+2}{i}^{-1} c_{\ell+1} w_{n+2}^{(i)}(\ell)$;
- $2\Delta\rho(\ell+1)f_1$ with component $i \le n$, by the second inequality of (5.1) and indices $(1, n-i)$, gives at least $2\binom{n}{i}\binom{n+2}{i+1}^{-1} c_{\ell+1} w_{n+2}^{(i+1)}(\ell)$;
- $P_2(\ell)f_0$ with component n gives at least $c_{\ell+1} w_{n+2}^{(n+2)}(\ell)$, since $P_2(\ell)P_n(\ell+1) \ge (P_0 + P_1)(\ell)P_{n+2}(\ell)$ by log-concavity in k, and $f_0(\ell+1) = d_{\ell+1}$.

When $n - i = 0$ the index condition is trivial. Collecting by target,

$$\varphi_2(\ell)\widehat{w}_n(\ell+1) \ge c_{\ell+1} \sum_{i=2}^{n+2} \theta_{n,i} w_{n+2}^{(i)}(\ell),$$

where $\theta_{n,2} = \binom{n}{2}\binom{n+2}{2}^{-1}$, $\theta_{n,i} = 1 - \frac{i(i-1)}{(n+1)(n+2)}$ for $3 \le i \le n$, $\theta_{n,n+1} = \frac{2}{n+2}$ and $\theta_{n,n+2} = 1$.

As in the proof of Theorem 4.2, the coefficient quotients are $\frac{j(m-1)(m-2)}{2(m-j-1)}$ for V, and $\theta_{m-2,i}\frac{jm(m-1)}{2(m-j-1)}$ for $\widehat{W}$, on $m \ge 2j+1$, $j \ge 2$. The first is at least 6, by $(m-4)(m-5) \ge 0$ at $j=2$. The second exceeds 2: it is at least $\frac{j(m-2)(m-3)}{2(m-j-1)} \ge m-2$ for $i=2$, at least $\frac{j(2m-3)}{m-j-1}$ for $3 \le i \le m-2$, equal to $\frac{j(m-1)}{m-j-1}$ for $i = m-1$, and at least 10 for $i = m$. Hence

$$B_0V'_{j-1} \ge 6c_{\ell+1}V_j \qquad \text{and} \qquad B_0W'_{j-1} \ge 2c_{\ell+1}W_j \qquad (j \ge 2).$$

Resummation. We prove below, for $(\ell,a)\in\Omega$ with $a\ge 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1)+B_0(\ell,a)A_0^{\psi}(\ell+1,a-2)\ \ge\ c_{\ell+1}\big(4H_1-2V_1\big)(\ell,a). \tag{5.2}$$

With the notation of the proof of Theorem 4.2, the inequalities above give $P^2\ge 4Q$, $H_+P\ge 10H_-$, $V_+P\ge 6V_-$, $W_+P\ge 2W_-$, $C_+P\ge 6C_-$ and $A_0P+PA_0+2V_1\ge 4H_1$. Since $(I-P/\mu)G=(1-\frac{2}{\mu})G+\frac{2}{\mu}I$, one has $G(H_+-H_-)G\ge 0$, $GCG\ge 0$, $G(W_+-W_-)G\ge 0$ and $G(V_+-V_-)G\ge\frac{2}{3}GV_1G$, so

$$GAG\ \ge\ G\big(A_0-\tfrac14(A_0P+PA_0)+\tfrac16V_1\big)G\ =\ \tfrac12(GA_0+A_0G)+\tfrac16GV_1G\ \ge\ 0,$$

and (P) follows as before.

Proof of (5.2): notation. Use the notation of the proof of Theorem 4.1: $u_n=f_n(\ell)$, which is log-concave without internal zeros, y_n, D, X, X', Y, Y', J, H_*. Put also $A=d_{\ell+a}$, $B=d_{\ell+a-1}$, $F=A-B=f_1(\ell+a-1)$, $X''=B-d_\ell$ and $P_k=P_k(\ell)$. Since $c_{\ell+n}=d_{\ell+n}\sum_k\binom{n}{k}P_k$ for $n\le a$, the following are linear in P:

- $B_0(\ell,a)=\sum_k\beta_kP_k$ and $B_0(\ell+1,a-1)=\sum_k\beta'_kP_k$, where

  $\beta_0=Y$, $\beta_1=a(A-D)$, $\beta_k=\binom{a}{k}A$ $(k\ge 2)$,
  $\beta'_0=Y'$, $\beta'_1=aA-2(a-1)d_{\ell+2}+(a-2)D$, $\beta'_2=\binom{a}{2}A-(a-1)d_{\ell+2}$, $\beta'_k=\binom{a}{k}A$ $(k\ge 3)$;

- $c_{\ell+a}-c_\ell=\sum_k\chi_kP_k$ and $c_{\ell+a-1}-c_{\ell+1}=\sum_k\chi'_kP_k$;
- $c_{\ell+a}-c_{\ell+a-1}=\sum_k\delta_kP_k$ with $\delta_k=\binom{a}{k}F+\binom{a-1}{k-1}B$;
- $H_1=H_*-\sum_k\zeta_kP_k$ with $\zeta_k=\sum_{n\ge3}(n-2)\binom{n}{k}y_n$, and $V_1=\sum_k\nu_kP_k$ with $\nu_k=\sum_{n\ge3}(n-2)n\binom{n-1}{k-1}\binom{a}{n}u_{n-1}$;
- $c_{\ell+1}=D(P_0+P_1)$.

As $A_0^{\psi}(\ell,a)=X-(c_{\ell+a}-c_\ell)$ and $A_0^{\psi}(\ell+1,a-2)=X'-(c_{\ell+a-1}-c_{\ell+1})$, the difference of the two sides of (5.2) is $\mathrm{Key}(P)=\mathrm{Lin}(P)-\mathrm{Quad}(P)$. Here $\mathrm{Lin}(P)=\sum_k l_kP_k$ with $l_k=X\beta'_k+\beta_kX'-4[k\le1]DH_*$, and Quad is a quadratic form. We use repeatedly:

$$AX''\ge(a-1)d_\ell F,\qquad AX'\ge(a-2)DF,\qquad u_0F\le u_1B,\qquad FD\le u_1A. \tag{5.3}$$

The first two hold because $A/F\ge R_{\ell+a-1}\ge R_{\ell+i}$ gives $(A/F)f_1(\ell+i)\ge d_{\ell+i}$ for $i\le a-1$; summing over $0\le i\le a-2$, respectively $1\le i\le a-2$, gives them. The last two follow from $u_0u_{m+1}\le u_1u_m$, as $F=\sum_m\binom{a-1}{m}u_{m+1}$, $B=\sum_m\binom{a-1}{m}u_m$ and $D=u_0+u_1$.

(i) Reduction to a quadratic form in P. By monotonicity of b at $\ell+a-1$, $\Delta c:=c_{\ell+a}-c_{\ell+a-1}\le F$. With $\Phi(P):=\Delta c\cdot\mathrm{Lin}(P)-F\,\mathrm{Quad}(P)$ one has $\mathrm{Key}=(1-\Delta c/F)\mathrm{Lin}+\Phi/F$, so it suffices that every $l_k\ge0$ and $\Phi\ge0$. With $\chi_k=\binom{a}{k}A-[k=0]d_\ell$ and $\chi'_k=\binom{a-1}{k}B-\binom{1}{k}D$, so that $E^1_m=X\delta_m-F\chi_m$, $E^2_m=X'\delta_m-F\chi'_m$ and $E^3_m=2F\nu_m-4(H_*\delta_m-F\zeta_m)$, expansion gives $\Phi(P)=\sum_{k,m}M_{k,m}P_kP_m$ with

$$M_{k,m}\ =\ \beta'_kE^1_m+\beta_kE^2_m+[k\le1]D\,E^3_m,$$

and explicitly $E_0^1 = E_0^2 = E_0^3 = 0$ and, for $m \ge 1$,

$$E_m^1 = \tbinom{a-1}{m-1}AX'' - \tbinom{a-1}{m}d_\ell F, \qquad E_m^2 = \tbinom{a-1}{m-1}AX' - \Big(\tbinom{a-1}{m} - [m=1]\Big)DF,$$
$$E_m^3 = 2F\nu_m - 4\tbinom{a-1}{m-1}BH_* - 4FZ_m, \qquad Z_m = \sum_{n\ge 3}(n-2)\Big(\tbinom{a}{m} - \tbinom{n}{m}\Big)y_n.$$

Since (P_k) is log-concave without internal zeros, the Abel summation in the proof of Lemma 5.4, applied to every anti-diagonal of Φ, shows $\Phi \ge 0$ if every central sum $S_{\sigma,i} = \sum_{k=i}^{\sigma-i} M_{k,\sigma-k}$, $0 \le i \le \sigma/2$, is nonnegative. As $M_{k,0} = 0$, for $\sigma \ge 3$

$$S_{\sigma,0} = S_{\sigma,1} + M_{0,\sigma} \qquad \text{and} \qquad S_{\sigma,1} = S_{\sigma,2} + M_{1,\sigma-1} + M_{\sigma-1,1}.$$

Since $\binom{a-1}{m} = \frac{a-m}{m}\binom{a-1}{m-1}$ and $\frac{a-m}{m(a-1)} \le \frac{a-m}{m(a-2)} \le \frac12$ for $m \ge 2$, (5.3) gives, for $m \ge 2$,

$$E_m^1 \ge \tfrac12\tbinom{a-1}{m-1}AX'' \ge 0 \qquad \text{and} \qquad E_m^2 \ge \tfrac12\tbinom{a-1}{m-1}AX' \ge 0.$$

Also every $\beta_k \ge 0$ and $\beta_k' \ge 0$; for β_2' use $\binom{a}{2}A \ge (a-1)d_{\ell+2}$. Finally $E_1^1 = AX'' - (a-1)d_\ell F \ge 0$ and $E_1^2 = AX' - (a-2)DF \ge 0$.

(ii) The central sums with $i \ge 2$, and the coefficients l_k. For $k, m \ge 2$, $M_{k,m} = \beta_k' E_m^1 + \beta_k E_m^2 \ge 0$, so $S_{\sigma,i} \ge 0$ for $i \ge 2$. Next, $l_0 = K \ge 0$ by the proof of Theorem 4.1, and $l_k \ge 0$ for $k \ge 2$ termwise. By convexity, $A - D \ge (a-1)f_1(\ell+1) = (a-1)(d_{\ell+2} - D)$, so $\beta_1' \ge (a-2)(A-D)$ and

$$l_1 \ge L := (A-D)\big((a-2)X + aX'\big) - 4DH_*.$$

(iii) The central sums with $\sigma \le 2$. Take $P = (P_0, P_1, 0, \dots)$ with $P_0 \ge 0$ and $P_1 > 0$, and put $\lambda = F/\Delta c(P)$. Then $\mathrm{Key}(\lambda P) = (\lambda/\Delta c)\Phi(P)$, and λP is the linear ratio $\tilde\rho(\ell + x) = \lambda(P_0 + P_1 x)$. Its $\tilde b = (1-\tilde\rho)d$ satisfies $\Delta\tilde b(\ell + a - 1) = 0$. So $\alpha := 1 - \tilde\rho(\ell + a) = tR_{\ell+a-1} \ge tu_0/u_1$ with $t = \lambda P_1$. For such α and t, the proof of Theorem 4.1 shows (4.2). Its two weights A_0^ψ are nonnegative, since $X \ge au_1$ and $X' \ge (a-2)u_1$, and $B_0(\ell, a) \ge \tilde\rho(\ell+1)Y$, $B_0(\ell+1, a-1) \ge \tilde\rho(\ell+1)Y'$, since $\beta_1 \ge Y$ and $\beta_1' \ge Y'$. Hence $\mathrm{Key}(\lambda P) \ge 0$, that is, $\Phi(P) = P_1(M_{0,1}P_0 + M_{1,1}P_1) \ge 0$. This gives $S_{1,0} = M_{0,1} \ge 0$ and $S_{2,1} = M_{1,1} \ge 0$. The sum $S_{2,0} = M_{0,2} + M_{1,1}$ is treated in (v).

(iii′) $M_{1,m} + M_{m,1} \ge 0$ for $m \ge 2$ and $a \ge 6$. Put $c_m' = \binom{a-1}{m-1}$. Then

$$M_{1,m} + M_{m,1} = \big[\beta_1 E_m^2 - 4c_m' DBH_*\big] + \big[\beta_1' E_m^1 - DF(4Z_m - 2\nu_m)\big] + \beta_m' E_1^1 + \beta_m E_1^2,$$

and the last two terms are nonnegative.

- First bracket: $E_m^2 \ge \frac12 c_m' AX'$ and $A \ge B$, so it is nonnegative if $R := a(A-D)X' - 8DH_* \ge 0$.
- Second bracket: $Z_m \le \binom{a}{m}H_* \le \frac{a}{2}c_m' H_*$, $\nu_m \ge 0$, $F \le A$, $E_m^1 \ge \frac12 c_m' AX''$ and $\beta_1' \ge (a-2)(A-D)$. So it is nonnegative if $R_2 := (a-2)(A-D)X'' - 4aDH_* \ge 0$.

(iv) $M_{0,s} \ge 0$ for $s \ge 3$. Put $c' = \binom{a-1}{s-1}$ and $c'' = \binom{a-1}{s}$. Then

$$M_{0,s} = Y'E_s^1 + YE_s^2 + DE_s^3 = c'BK'' + FG_s,$$

where

$$K'' = Y'X'' + YX' - 4DH_*, \qquad G_s = c'(Y'X'' + YX') - c''(d_\ell Y' + DY) + D(2\nu_s - 4Z_s).$$

Note $u_1 = d_{\ell+1} - d_\ell > 0$. If $K'' \ge 0$, then $u_1 B \ge u_0 F$ gives $u_1 M_{0,s} \ge F(c'u_0 K'' + u_1 G_s)$, and

$$c'u_0K'' + u_1G_s = D\,Q_s + c''u_1^2Y', \qquad Q_s = c'(Y'X'' + YX') - 4c'u_0H_* + u_1(2\nu_s - 4Z_s) - c''u_1(Y + Y').$$

So $M_{0,s} \ge 0$ once $K'' \ge 0$ and $Q_s \ge 0$.

(v) $S_{2,0} \ge 0$ for $a \ge 5$. With $c'' = \binom{a-1}{2}$, the definitions give the identity

$$S_{2,0} = \beta_1' E_1^1 + \big[\beta_1 E_1^2 - c''F(d_\ell Y' + DY)\big] + B\,K_a + F\,W,$$

where $K_a = (a-1)K'' - 4DH_*$ and $W = (a-1)(Y'X'' + YX') + D(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

- Since $\beta_1 = a(X' + F)$, $E_1^2 = AX' - (a-2)DF$, $d_\ell \le D$ and $FD \le u_1 A$, the bracket is at least $A\Psi$, where $\Psi = a(A - D)J - c''u_1(Y + Y')$. Recall $J = X' - (a-2)u_1$ and $X' + F = A - D$.
- If $K_a \ge 0$, then $u_1(BK_a + FW) \ge F(u_0K_a + u_1W) = FD\widehat{Q}$, where $\widehat{Q} = (a-1)(Y'X'' + YX') - 4au_0H_* + u_1(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

Hence $u_1 S_{2,0} \ge u_1\beta_1' E_1^1 + u_1 A\Psi + FD\widehat{Q} \ge 0$ once K_a, Ψ and $\widehat{Q}$ are nonnegative.

(vi) Edge bounds. Steps (ii)–(v) reduce (5.2) to the nonnegativity of the quadratic forms K, L, R, R_2, K'', Q_s $(3 \le s \le a)$, K_a, Ψ and $\widehat{Q}$ in the sequence u. For Q_s, first remove the dependence on s: by $\nu_s \ge 0$, $Z_s \le \binom{a}{s}H_*$, $\binom{a}{s} \le \frac{a}{3}\binom{a-1}{s-1}$ and $\binom{a-1}{s} \le \frac{a-3}{3}\binom{a-1}{s-1}$ for $s \ge 3$, one has $Q_s \ge \binom{a-1}{s-1}Q^*$ with

$$Q^* = Y'X'' + YX' - 4u_0H_* - \tfrac{4a}{3}u_1H_* - \tfrac{a-3}{3}u_1(Y + Y').$$

Each of these forms is a sum of products xy of linear forms $x = \sum_{i\ge1} x_i u_i$ and $y = \sum_{j\ge1} y_j u_j$ with nonnegative coefficients, plus u_1 times a linear form, minus u_0 times a linear form with nonnegative coefficients. Since u is log-concave without internal zeros, $u_iu_j \ge u_1u_{i+j-1}$ for $i, j \ge 1$ and $u_0u_{n+1} \le u_1u_n$. Keeping in xy only the pairs with $i \le 2$ or $j \le 2$ gives

$$xy \ge u_1 \sum_n e_n(x,y)u_n, \qquad e_n(x,y) = x_1y_n + x_2y_{n-1} + [n \ge 3]\,y_1x_n + [n \ge 4]\,y_2x_{n-1},$$

and $u_0 \sum_{n\ge1} h_n u_n \le u_1 \sum_{n\ge0} h_{n+1}u_n$. So each form is at least $u_1 \sum_n e_n u_n$ with explicit coefficients e_n, and it suffices that every $e_n \ge 0$.

- For $4 \le n \le a$ write $n = 4+p$ and $a = n+k$. Then $e_n/\binom{a}{n}$ is a rational function of (p,k) whose numerator has nonnegative coefficients and whose denominator is strictly positive for $p,k \ge 0$.
- For $n \le 3$, e_n is a polynomial in a whose coefficients are nonnegative after $a \mapsto a+a_0$, with $a_0 = 6$ for R_2, $a_0 = 5$ for K_a and $a_0 = 3$ otherwise.
- For $n \ge a+1$ only nonnegative products remain.

These certificates are formalized in Lean, and `runs/lean_plan/edge_exact.py` checks every e_n in exact arithmetic for $a \le 60$.

(vii) Small windows. Steps (iii′) and (v) need $a \ge 6$ and $a \ge 5$. For $3 \le a \le 5$, each sum $M_{1,m}+M_{m,1}$ $(2 \le m \le a)$ and, for $a \le 4$, $S_{2,0}$ is a cubic form in u that is an exact nonnegative rational combination of the products $u_k(u_{i+1}u_{j-1}-u_iu_j)$ $(i+1 \le j-1)$ and of monomials (`runs/lean_plan/window_cert.py`). This proves (5.2) and the theorem. ∎

For cubic ratios, log-concavity of the Newton coefficients can be replaced by a single inequality.

Theorem 5.6 *(Cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t,\tau,\sigma \ge 0$ and $r\sigma \le t\tau$. Put $a_i = 1-d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

If (r,t,τ,σ) is log-concave without internal zeros, then $r\sigma \le t\tau$: for $t\tau > 0$ this follows from $t^2\tau^2 \ge r\tau \cdot t\sigma$, and otherwise $\sigma = 0$. Theorem 5.6 also admits cubic ratios with $t^2 < r\tau$ or $\tau^2 < t\sigma$, which Theorem 5.5 does not cover. The failing cubic examples of Section 6 violate $r\sigma \le t\tau$. The proof is computer-assisted: the inequalities (5.5), (5.6) and (5.4) and the inequality $\Phi_3^{\mathrm{loc}} \ge 0$ below are reduced to families of explicit polynomials with nonnegative coefficients.

Proof. Setup. As in the proof of Theorem 4.2, $R_i = d_i/f_1(i)$ is nondecreasing, $1-\rho(m+1) \ge \Delta\rho(m)R_m$ for $0 \le m < N$, and ρ is nondecreasing with $0 < \rho(i) \le 1$ for $i \le N$; here $\Delta\rho(x) = t + \tau x + \sigma\binom{x}{2}$ and $\Delta^2\rho(x) = \tau + \sigma x$. For $\ell + n \le N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \tbinom{n}{2}\Delta^2\rho(\ell+n-2)f_{n-2} + \tbinom{n}{3}\sigma f_{n-3}.$$

Since $\binom{\ell+n-1}{2} = \binom{\ell}{2} + \ell(n-1) + \binom{n-1}{2}$ and $n\binom{n-1}{2} = \binom{n}{2}(n-2) = 3\binom{n}{3}$, this gives $\psi_n = h_n - v_n - e_n$ with $h_n = (1-\rho(\ell+n))f_n$,

$$v_n = n\Gamma_n(\ell)f_{n-1}, \quad \Gamma_n(\ell) = t + \tau(\ell+n-1) + \sigma\Big(\tbinom{\ell}{2} + \ell(n-1)\Big), \qquad e_n = \tbinom{n}{2}(\tau+\sigma\ell)f_{n-2} + \sigma\tbinom{n}{3}\big(3f_{n-1}+3f_{n-}$$

all nonnegative. Write $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$; then (u_k) is log-concave, and $u_0 \le \ell u_1$ by convexity.

Moves. Split the moves as in the proof of Theorem 4.2, with V_j formed from v_n and with a third family $E_j = W_{2j+1,j}$ formed from e_n in place of W_j.

Detours of H, V and C'. The detours of H and C' hold verbatim, with constants 10 and 6. For V,

$$(n+1)\Gamma_n(\ell+1) - n\Gamma_{n+2}(\ell) \;=\; t + \tau\ell + \sigma\Big(\tbinom{\ell}{2} + n^2 - 1\Big) \;\geq\; 0,$$

so Lemma 2.1 with indices $(2, n-1)$ gives $(n+1)(n+2)\varphi_2 v_n' \geq n^2 c_{\ell+1} v_{n+2}$ for $n \geq 3$, and $B_0 V_{j-1}' \geq \frac{9}{2} c_{\ell+1} V_j$ for $j \geq 2$ as before. (With the full σ-part $n\sigma\binom{\ell+n-1}{2} f_{n-1}$ in v_n, the corresponding difference is negative for $\ell < n+1$; without any σ-part in v_n, the reduction of the key inequality below fails for large ℓ.)

Detour of E. We show, for $n \geq 3$ and $\ell + n + 2 \leq N$,

$$(n+1)(n+2)\Big(\varphi_2(\ell) + \frac{n}{3}\varphi_3(\ell)\Big) e_n(\ell+1) \;\geq\; 3(n-1)\, c_{\ell+1} e_{n+2}(\ell). \tag{5.4}$$

Granting this, $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell) + \binom{a}{3}\varphi_3(\ell)$, $\binom{a}{3}\binom{a-2}{n} = \frac{a-2}{3}\binom{a}{2}\binom{a-2}{n}$ with $a - 2 \geq n$ whenever $\binom{a}{n+2} \neq 0$, and $\binom{a}{2}\binom{a-2}{n} = \binom{a}{n+2}\binom{n+2}{2}$ give, for $j \geq 2$,

$$B_0(\ell, a) E_{j-1}(\ell+1, a-2) \;\geq\; \tfrac{3}{2} c_{\ell+1} \sum_{m \geq 2j+1} (m-3)\tbinom{m-j-2}{j-1}\tbinom{a}{m} e_m(\ell) \;\geq\; 3 c_{\ell+1} E_j(\ell, a),$$

because $(m-3)\binom{m-j-2}{j-1} - 2\binom{m-j-1}{j} = \binom{m-j-2}{j-1}\frac{(j-2)(m-1)}{j} \geq 0$.

Detour of B. We show, for $n \geq 2$, $\ell + n + 2 \leq N$ and $(\ell, n) \neq (0, 2)$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \;\geq\; 8(n-1)\, c_{\ell+1}\varphi_{n+2}(\ell), \tag{5.5}$$

and, if $N \geq 4$,

$$\big(12\varphi_2(0) + 8\varphi_3(0)\big)\varphi_2(1) \;\geq\; 8 c_1 \varphi_4(0). \tag{5.6}$$

Granting these, the argument after (4.3) gives $B_0(\ell, a) B_{j-1}(\ell+1, a-2) \geq 4 c_{\ell+1} B_j(\ell, a)$ for $j \geq 1$. The only exception is the term $m = 4$ for $\ell = 0$, $j = 1$. There $B_0(0, a) \geq \binom{a}{2}\varphi_2(0) + \binom{a}{3}\varphi_3(0)$, $\binom{a}{2}\binom{a-2}{2} = 6\binom{a}{4}$ and $\binom{a}{3}\binom{a-2}{2} = 2(a-2)\binom{a}{4} \geq 4\binom{a}{4}$ reduce the term to (5.6). Inequality (5.5) fails at $(\ell, n) = (0, 2)$ in general.

Proof of (5.5), (5.6) and (5.4). In each, the difference of the two sides is a quadratic form $\sum \kappa_{\beta\gamma}\rho_\beta\rho_\gamma$ in $(\rho_0, \rho_1, \rho_2, \rho_3) = (r, t, \tau, \sigma)$. Its coefficients are bilinear forms in the u_k, with coefficients polynomial in ℓ and n.

- Since $t\tau\kappa_{t\tau} + r\sigma\kappa_{r\sigma} = (t\tau - r\sigma)\kappa_{t\tau} + r\sigma(\kappa_{t\tau} + \kappa_{r\sigma})$ and $r\sigma \leq t\tau$, the form is nonnegative if $\kappa_{t\tau}$, $\kappa_{t\tau} + \kappa_{r\sigma}$ and all coefficients other than $\kappa_{r\sigma}$ are nonnegative as forms in u.

- For (5.5) the coefficients κ_{rr}, $\kappa_{r\tau}$, $\kappa_{\tau\tau}$ need not be checked: the form with $t = \sigma = 0$ is (4.3) for the ratio $r + \tau\binom{x}{2}$. The same holds for (5.6), whose left side exceeds that of (5.5) at $(0, 2)$ by $8\varphi_3(0)\varphi_2(1) \geq 0$.

- Every remaining coefficient is a sum $\sum c_{ij} u_i u_j$ over $i \leq j$. Group its terms by $i + j = m$. Within a group, $u_i u_{m-i}$ is nondecreasing in i by log-concavity, so with the tail sums $T_k = \sum_{i \geq k} c_{i,m-i}$ and the smallest index k_0 of the group,

  $$\sum_i c_{i,m-i} u_i u_{m-i} \;=\; T_{k_0} u_{k_0} u_{m-k_0} + \sum_{k > k_0} T_k\,(u_k u_{m-k} - u_{k-1} u_{m-k+1}),$$

  and the coefficient is nonnegative if all $T_k \geq 0$.

- The tail sums are polynomials in (ℓ, n). For (5.5) with $n \geq 6$ and for (5.4) with $n \geq 7$ they have nonnegative coefficients in $(\ell, n-6)$, respectively $(\ell, n-7)$. For each n in $3 \leq n \leq 5$ ((5.5)) and $3 \leq n \leq 6$ ((5.4)) they are polynomials in ℓ with nonnegative coefficients. The same holds for (5.5) with $n = 2$ and $\ell \geq 1$ after $\ell \mapsto \ell + 1$, and for (5.6), where $u_0 = 0$.

The scripts `runs/lean_plan/gen_cubicB.py` and `runs/lean_plan/gen_cubicE.py` check every tail-coefficient sign in exact rational arithmetic and generate the Lean certificates; Lean checks the certificate identities.

Key inequality. We show, for $(\ell, a) \in \Omega$ with $a \geq 3$,

$$A_0^{\psi}(\ell, a) B_0(\ell+1, a-1) + B_0(\ell, a) A_0^{\psi}(\ell+1, a-2) \geq c_{\ell+1}\big(4H_1 - 2V_1 - E_1\big)(\ell, a). \quad (5.7)$$

Use the notation of the proof of Theorem 4.2, and put $P_1 = \Delta\rho(\ell)$ and $P_2 = \Delta^2\rho(\ell) = \tau + \sigma\ell$, so that $\rho(\ell + k) = \rho(\ell) + kP_1 + \binom{k}{2}P_2 + \binom{k}{3}\sigma$. Put $\alpha = 1 - \rho(\ell + a)$, $s_q = P_1 + (a-1)P_2$ and $s = \Delta\rho(\ell + a - 1) = s_q + \binom{a-1}{2}\sigma$. Then $\alpha \geq su_0/u_1$ and $s_q \geq (a-1)P_2$. Backward differences give

$$\begin{aligned} A_0^{\psi}(\ell, a) &= \alpha X - s_q a u_0 + P_2\tbinom{a}{2}u_0 - \sigma\tbinom{a}{3}u_0, \\ A_0^{\psi}(\ell+1, a-2) &= (\alpha + s)X' - s_q(a-2)D + P_2\tbinom{a-1}{2}D - \sigma\tbinom{a-1}{3}D, \end{aligned}$$

and, reindexing as before, $4H_1 - 2V_1 - E_1 = 4\alpha H_* + 2s_q T + 4P_2 Z - P_2\sum_{n\geq 1} n\binom{a-n}{2}y_n + \sigma(4T_3 - E_3)$, where $T_3 = \sum_{n\geq 3}(n-2)\big(\binom{a}{3} - \binom{n}{3}\big)y_n$ and $E_3 = \sum_{n\geq 3}(n-2)\binom{n}{3}\binom{a}{n}(3u_{n-1} + 3u_{n-2} + u_{n-3})$. With $B_0(\ell, a) \geq pY$, $B_0(\ell+1, a-1) \geq pY'$ and $c_{\ell+1} = pD$, it suffices that $R = A_0^{\psi}(\ell, a)Y' + YA_0^{\psi}(\ell+1, a-2) - D(4H_1 - 2V_1 - E_1) \geq 0$. R is affine in α, with coefficient $XY' + YX' - 4DH_*$. Its part without σ is the expression bounded in the proof of Theorem 4.2, with s_q and P_2 in place of s and τ; that bound holds for every value α' of α with $\alpha' u_1 \geq s_q u_0$. Apply it with $\alpha' = \alpha - \binom{a-1}{2}\sigma z$, where $z = u_0/u_1$. This gives $R \geq \sigma\Phi_3^{\mathrm{loc}}$ with

$$\Phi_3^{\mathrm{loc}} = \tbinom{a-1}{2}z\,(XY' + YX' - 4DH_*) - \tbinom{a}{3}u_0 Y' + Y\Big(\tbinom{a-1}{2}X' - \tbinom{a-1}{3}D\Big) - D(4T_3 - E_3).$$

$\Phi_3^{\mathrm{loc}} \geq 0$. Write $E_3 = \binom{a}{3}u_0 + E_3^0$ and $H_*^+ = \sum_{n\geq 2}(n-1)\binom{a}{n+1}u_n$, so that $u_0 H_* \leq u_1 H_*^+$. Bounding the only negative term quadratic in u_0, $-4\binom{a-1}{2}u_0^2 H_*$, this way and dropping $\binom{a}{3}u_0^2 u_1 \geq 0$ gives $u_1\Phi_3^{\mathrm{loc}} \geq u_1 A + u_0 B$ with

$$\begin{aligned} A &= \tbinom{a-1}{2}YX' - u_1\Big(\tbinom{a-1}{3}Y + 4T_3 - E_3^0\Big), \\ B &= \tbinom{a-1}{2}\big(XY' + YX' - 4u_1 H_* - 4u_1 H_*^+\big) - u_1\Big(\tbinom{a}{3}Y' + \tbinom{a-1}{3}Y + 4T_3 - E_3^0 - \tbinom{a}{3}u_1\Big). \end{aligned}$$

These are forms of the kind treated in step (vi) of the proof of Theorem 5.5: the edge bounds give $A \geq u_1\sum_n e_n^A u_n$ and $B \geq u_1\sum_n e_n^B u_n$. Their coefficients are nonnegative by the same three kinds of certificates: for $4 \leq n \leq a$ the numerators in (p, k) have nonnegative coefficients and the denominators are strictly positive for $p, k \geq 0$, for $n \leq 3$ the polynomials in a have nonnegative coefficients after $a \mapsto a + 3$, and for $n \geq a + 1$ only nonnegative terms remain (`runs/lean_plan/phi3_generic.py`, `runs/lean_plan/phi3_edge.py`). Hence $R \geq 0$.

Resummation. Proceed as in the proof of Theorem 4.2, with E in place of W. The inequalities $P^2 \geq 4Q$, $H_+P \geq 10H_-$, $V_+P \geq \frac{9}{2}V_-$, $E_+P \geq 3E_-$, $C_+P \geq 6C_-$ and $A_0P + PA_0 + 2V_1 + E_1 \geq 4H_1$ give $GAG \geq \frac{1}{2}(GA_0 + A_0G) + \frac{1}{18}GV_1G + \frac{1}{12}GE_1G \geq 0$, and (P) follows. ∎

The condition $r\sigma \leq t\tau$ enters only through (5.5) and (5.4), each time through $\kappa_{t\tau} + \kappa_{r\sigma} \geq 0$. On geometric profiles at $(\ell, n) = (0, 3)$ one has $\kappa_{r\sigma} = -20q(4q^3 + 11q^2 + 9q + 1)$ and $\kappa_{t\tau} = 20(q+1)(8q^3 + 33q^2 + 30q + 6)$ in (5.5), so $\kappa_{t\tau}/|\kappa_{r\sigma}| \to 2$ as $q \to \infty$. This geometric family therefore obstructs the coefficientwise test for $r\sigma \leq u\,t\tau$ with $u > 2$; it does not establish a cutoff at $u = 1$. In fact (5.5) and (5.4) also hold with $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma} \geq 0$ in place of $\kappa_{t\tau} + \kappa_{r\sigma} \geq 0$; the other coefficients and the key inequality do not involve the condition. For (5.4), and for (5.5) with $n \geq 6$, the tail sums above stay nonnegative. For (5.5) with $n \leq 5$ the transfers alone do not suffice. There the modified coefficient $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma}$ is certified after the substitution $u_0 = zu_1$, $0 \leq z \leq \min(\ell, 1/q)$ with $q = u_2/u_1$, in the regimes $z = 0$, $z = \ell$ and $z = 1/q$ (`runs/cubic/bdetour_u.py`, `runs/cubic/edetour_u.py`). The relevant expressions are affine in z, so these boundary regimes cover the admissible interval for $q > 0$. When $q = 0$, all u_k with $k \geq 2$ vanish, and the inequality follows directly. For (5.6), with $u_0 = 0$, the modified tail sums are also nonnegative. Here `runs/cubic/bdetour_u.py` supplies the small-n B certificate; `runs/cubic/edetour_u.py` separately certifies E. Hence Theorem 5.6 holds with $r\sigma \leq \frac{7}{4}t\tau$ in place of $r\sigma \leq t\tau$. This extension is machine-verified (`Szkraw.cubic_ratio74_matrix`). For (5.5) with $n \leq 5$ the Lean proof does not substitute $u_0 = zu_1$: it writes the modified coefficient, times a positive polynomial in ℓ, as a combination of transfers, products $u_a u_b$ and the terms $u_j(\ell u_1 - u_0) \geq 0$ with multipliers that are polynomials in ℓ with nonnegative coefficients after a shift of ℓ, plus exact rational certificates at the few smaller values of ℓ (`runs/lean_plan/cert74.py`, `runs/lean_plan/gen_cubicB.py`). With $\frac{9}{5}$ the certificate of `runs/cubic/bdetour_u.py` fails. Exact diagnostics for this proof are listed in `papers/DIAGNOSTICS.md`.

6. Counterexamples and open problems

The criterion of [1] retains the full Krawtchouk range with a constant ratio b_i/c_i. Theorem 3.1 and Corollary 3.2 allow the ratio $c_i/(b_i + c_i)$ to increase, provided the Newton coefficients of c are log-concave without internal zeros and those of b are those of c with nonincreasing nonnegative weights, Proposition 3.3 covers linearly increasing ratios with a linear profile, and Theorem 4.1 covers linearly increasing ratios with log-concave Newton profiles and nondecreasing b, Theorem 4.2 extends this to increasing convex quadratic ratios, and Theorems 5.5 and 5.6 to ratios of higher degree. Find direct inequalities that cover other varying ratios, for instance nonlinear increasing ratios, which Theorems 3.1 and 4.1 do not cover in general. Log-concavity of the Newton coefficients of both d and c together with monotone b does not suffice: $d = (0, 25, 55, 91)/91$, $c = (0, 24, 54, 90)/91$ gives $g(3, 3; 3) = -3/2002$. Increasing ratios with nondecreasing b can fail as well: for the linear profile $d_i = i/3$ and the concave ratio $\rho = (1/2, 3/4, 5/6)$, for which $b_i = 1/6$ for $i \geq 1$, one has $g(3, 3; 3) = -1/24$, and a convex kinked ratio on the linear profile fails at $N = 120$ (the example below with

$r = 2/5$ and $\rho_N = r + (1-r)/N$). Theorem 4.2 does not extend to cubic ratios: for $N = 4$, $d_i = i/4$ and $\rho(x) = \frac{19}{20} + \binom{x}{3}/320$, whose Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$ are nonnegative, $b = (0, \frac{1}{80}, \frac{1}{40}, \frac{9}{256}, \frac{3}{80})$ is nondecreasing, but $g(4,4;4) = -14011/30720000$. Even the bound $r \le \frac{9}{10}$ does not restore (P) for cubic ratios: on linear profiles with $\kappa = 1/80$, which keeps b nondecreasing, $\rho_i = \frac{9}{10} + \kappa\binom{i}{3}/\binom{N}{3}$ gives $g(48,48;48) < 0$ for $N = 48$, and $\rho_i = \frac{9}{10} + \kappa(i/N)^3$ gives $g(64,64;63) < 0$ for $N = 64$. Steeper ratios fail at smaller r, again with nondecreasing b and nonnegative Newton coefficients of ρ: $\rho_i = \frac{1}{2} + 10^{i-N}/100$ gives $g(40,40;40) < 0$ for $N = 40$, and raising only the last ratio, $\rho_i = r$ for $i < N$ and $\rho_N = r + (1-r)/N$, gives $g(40,40;38) < 0$ for $r = \frac{1}{2}$, $N = 40$ and $g(120,120;119) < 0$ for $r = \frac{2}{5}$, $N = 120$ (`runs/varratio/frontier_examples.py`). Write a cubic ratio as $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The cubic examples above violate $r\sigma \le t\tau$, a consequence of log-concavity without internal zeros of (r,t,τ,σ): in the first two $t = \tau = 0$, and for $\kappa(i/N)^3$ one has $t\tau = 6\kappa^2/N^6 < r\sigma = 6r\kappa/N^3$. In sampled arrays with $t, \tau \ge 0$, $r\sigma = u\,t\tau$ and nondecreasing b on log-concave Newton profiles, no negative coefficients were found for the tested values $u \in \{\frac{1}{4}, 1, 4, 16, 64\}$, $r \in \{\frac{1}{10}, \frac{1}{3}, \frac{1}{2}, \frac{4}{5}, \frac{19}{20}, \frac{99}{100}\}$ and $4 \le N \le 18$ (full tables), and in the coefficients $g(m, N; k)$ for $u \in \{1, 4, 16\}$ and $N = 24, 32, 48, 64, 96$ (`runs/varratio/cubic_lc_test.py`); in these samples the operator M of the proof of Theorem 4.1 was nonnegative for $u \le 4$ and $N \le 13$. For $t = \tau = 0$, by contrast, M has negative entries; for instance $N = 6$, $d_i = i/6$, $r = \frac{1}{5}$ and $\sigma = \frac{1}{200}$ give a negative entry (`runs/varratio/cubic_diag.py`); this obstructs extending that resummation to such data, and does not contradict Theorem 4.1, which does not cover them. Theorem 5.6 and the remark after its proof show that $r\sigma \le u\,t\tau$ suffices for $u \le \frac{7}{4}$. Decide whether it suffices for larger u; the proof of Theorem 5.6 uses the condition only coefficientwise, and on geometric profiles that test fails for $u > 2$. Theorem 5.5 admits ratios with log-concave Newton coefficients, of any degree, on every log-concave Newton profile; its proof is computer-assisted. Find a proof of its key inequality (5.2) without computer certificates, and decide whether, for ratios of degree greater than three, log-concavity of the ratio's Newton coefficients can be weakened to conditions like the one in Theorem 5.6. The terminating row and products of degree sum greater than N must remain part of the conclusion.

AI-use disclosure

OpenAI Codex and Anthropic Claude were used to develop proof arguments, check source passages, construct exact-arithmetic diagnostics, draft and revise this manuscript, and prepare the Lean formalization. The resummation in the proof of Theorem 4.1, its key inequality under nondecreasing b, and the weighted detour and key inequality in the proof of Theorem 4.2 arose in research consultations with ChatGPT Pro; the exact checks were completed with Claude, which also prepared the Lean formalizations. Theorem 5.1, Corollary 5.2, Theorems 5.5 and 5.6 and Lemma 5.4 were found and proved with Claude, without research consultation; Proposition 5.3, which is due to Walkup, was rederived with Claude together with the self-contained proof given here. Every result was reviewed by ChatGPT Pro, as part of a longer manuscript (an AI review with independent exact checks, not human peer review); the review rounds and their corrections are recorded in `REVIEW.md`, and each

returned PASS or PASS WITH MINOR FIXES, with the fixes applied. For Theorems 5.5 and 5.6 the reviewer reran transcribed copies of the certificate scripts; the scripts were also run unmodified from a fresh clone of the companion code at the cited revision, and all of them passed (`runs/verify/RESULTS.md`). The author set the research objective and retains responsibility for publication decisions and the final mathematical claims. No independent human verification is asserted here. "Machine-verified" refers only to the Lean theorems listed in the section on code and formalization; Theorems 5.5 and 5.6 and the extension to $r\sigma \leq \frac{7}{4}t\tau$ in the remark after the proof of Theorem 5.6 are machine-verified.

Code and formalization

The companion code [6] contains the certificate scripts cited in the proofs of Theorems 5.5 and 5.6, exact-arithmetic diagnostics for every result (`papers/DIAGNOSTICS.md`), and a Lean 4 development, built and audited by `make lean` and `make axioms`; the audit rejects dependencies outside `propext`, `Classical.choice` and `Quot.sound`. The declarations below have compiled proofs without added axioms or admitted goals. Notes on how each Lean statement matches the text are in `papers/FORMALIZATION.md`; this file, `papers/DIAGNOSTICS.md` and `runs/verify/RESULTS.md` are in the later revision [7].

Result	Lean declaration
Theorem 3.1: varying ratio	`Szkraw.varyratio_matrix`
Corollary 3.2: nonincreasing weights	`Szkraw.varyratio_weights`
Proposition 3.3: quadratic arrays	`Szkraw.quadratic_matrix`
Theorem 4.1: linearly increasing ratio	`Szkraw.affine_matrix`, `Szkraw.affine_bmono_iff`
Remarks after Theorem 4.1	`Szkraw.affine_range_remarks`, `Szkraw.profile_last_le`, `Szkraw.affEx_remark`, `Szkraw.affine_linear_quadratic`
Theorem 4.2: convex quadratic ratio	`Szkraw.quadratic_ratio_matrix`, `Szkraw.quadratic_bmono_iff`
Remarks on Theorem 4.2	`Szkraw.quadExample_remark`, `Szkraw.quadB_unit_fails`
Theorem 5.1: low-degree profiles	`Szkraw.lowdeg_matrix`
Corollary 5.2	`Szkraw.lowdeg_newton_lc`, `Szkraw.lowdeg_ratio_matrix`
Proposition 5.3: products of Newton sequences	`Szkraw.newton_product_LCNZ`
Lemma 5.4	`Szkraw.abel_weight`
Theorem 5.5: log-concave ratio	`Szkraw.lc_general`
Theorem 5.6: cubic ratio	`Szkraw.cubic_ratio_matrix`
Remark after the proof of Theorem 5.6: extension to $r\sigma \le \frac{7}{4}t\tau$	`Szkraw.cubic_ratio74_matrix`
Section 6: the counterexamples with $N = 3$ and the cubic counterexample with $N = 4$	`Szkraw.openEx1_fails`, `Szkraw.openEx2_fails`, `Szkraw.cubic_ratio_fails`
Section 6: the counterexamples with $N = 48$, 64, 40, 40 and 120	`Szkraw.cubicBinom48_fails`, `Szkraw.cubicPow64_fails`, `Szkraw.steepRatio40_fails`, `Szkraw.topRatio40_fails`, `Szkraw.topRatio120_fails`
Remarks after Corollary 3.2	`Szkraw.strict_ratio_example`, `Szkraw.openEx1_b_newton`, `Szkraw.linear_ratio_concave_b`
After Proposition 3.3: each hypothesis is needed	`Szkraw.quadratic_hypotheses_needed`
Section 5: exponential ratios, the ratio of the cubic counterexample, containments and noncomparability	`Szkraw.exponential_ratio`, `Szkraw.cubic_example_ratio_newton`, `Szkraw.general_contains`, `Szkraw.quad_lowdeg_incomparable`
After Lemma 5.4: weight sums on geometric sequences	`Szkraw.weight_sums_geometric`
Proof of Theorem 5.5: $(\log\rho)''(0) > 0$ for $\rho(x) = 1 + x/2 + x^2/2$	`Szkraw.log_rho_not_concave`
Remark after the proof of Theorem 5.6: geometric profiles and the case $u > 2$	`Szkraw.geometric_kappa_values`, `Szkraw.geometric_kappa`
Section 6: a negative entry of the operator M for $N = 6$, $t = \tau = 0$	`Szkraw.opM_negative_entry`, `Szkraw.opM_has_negative_entry`
After Proposition 5.3: the law of $\lvert S \cup T\rvert$, ultra-log-concavity of $\lvert S \cup T\rvert$, and the double-binomial convolution $h_k = \kappa_k$	`Szkraw.RandomSets.union_card_law`, `Szkraw.RandomSets.random_sets_ulc`, `Szkraw.RandomSets.walkup_convolution`

Not formalized: the identification of the random-set reading after Proposition 5.3 with Walkup's formulation (his conditionally uniform random sets and their fundamental sequences), which concerns the literature; the reported failure of `runs/cubic/bdetour_u.py` at $u = 9/5$, which concerns one certificate search, not property (P); and, in Section 6, the sampled tests of cubic ratios and of the resummed operator, which are numerical evidence over finitely many arrays. More generally, attributions to other works and statements about what they contain are not formalized, for instance that Proposition 5.3 is Walkup's Theorem 2 in the Newton basis and that the Abel argument is that of Wang and Yeh: Lean checks our statements, not the cited papers.

References

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[6] L. P. Polzer, *Companion code for finite nonnegative linearization* (2026), revision `2f96f4d` (full identifier in the URL), https://github.com/skypher/szkraw/tree/2f96f4d228215b35e31729f1d1526cd31879f161.

[7] L. P. Polzer, *Companion code for finite nonnegative linearization*, revision `4b2d7e1` with the split documentation `papers/FORMALIZATION.md`, `papers/DIAGNOSTICS.md` and the fresh-clone record `runs/verify/RESULTS.md` (full identifier in the URL), https://github.com/skypher/szkraw/tree/4b2d7e13d5c9a9fe8bbd406959d0ecd2cc192de8.